

GIS: Geographic Information Systems

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Outline

- Applications of GIS in Cultural Heritage
- The basic concepts of GIS
 - GIS as layers
 - Vectors and rasters
 - Coordinate reference systems
 - Symbology
- **Demo:** Getting started with QGIS
- **Demo:** least-cost analysis in QGIS

GIS in Cultural Heritage

Applications of GIS in Cultural Heritage

- Site discovery
- Crowdsourcing heritage data
- Cultural Heritage data management, monitoring, and preservation
- Analysis of historical locations and documents
- Modelling of mobility and circulation patterns
- Modelling of spatially rich documents (texts, archives, maps)
- In combination with other applications:
 - Remote sensing
 - Aerial and satellite imagery
 - Predictive and agent-based models
 - 3D models and Virtual Reality
 - Etc...

Site Discovery

- Remote-sensing techniques can be used to document the location of archaeological and historical sites and identify standing features

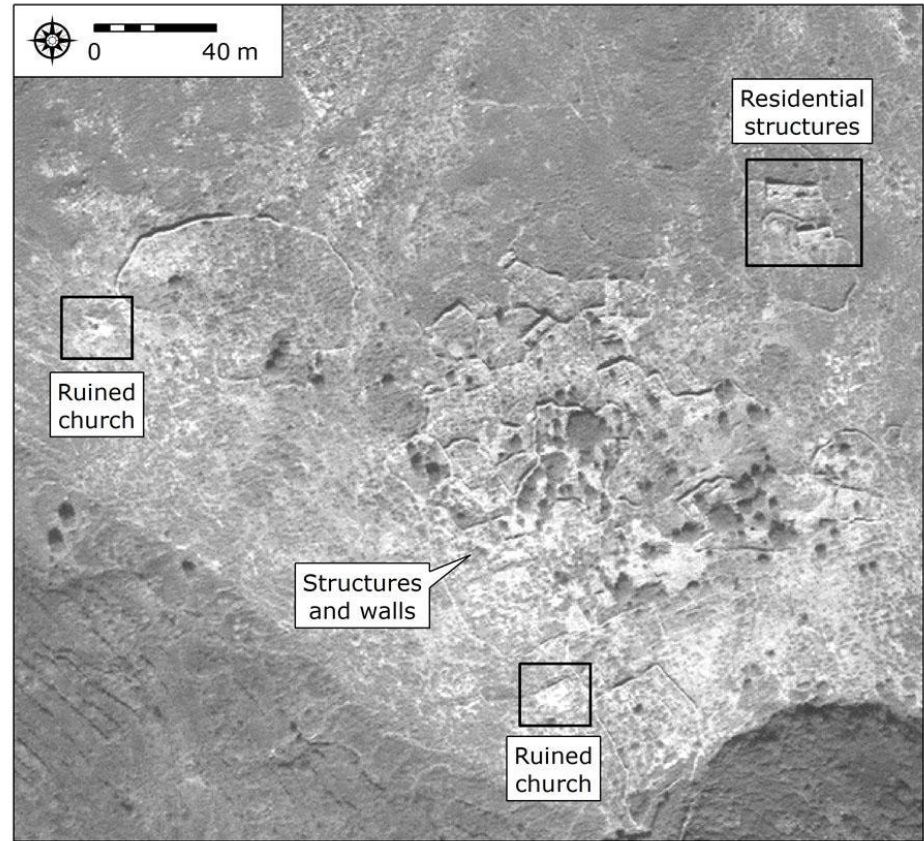
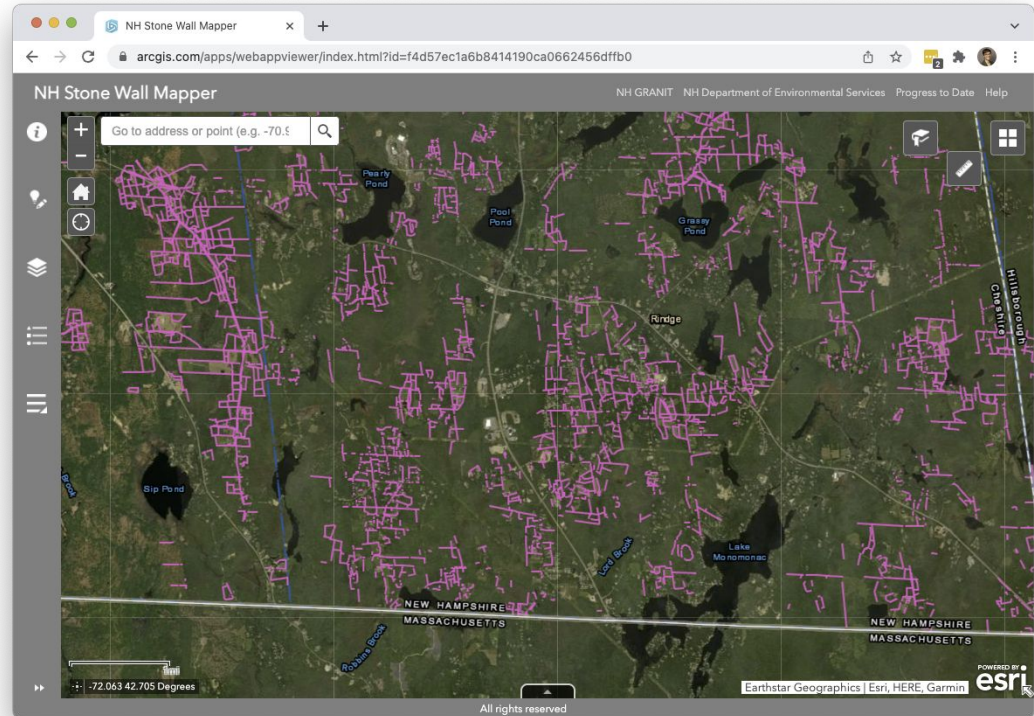


Figure 39. Site T360 (Proskephalia) identified in WorldView-2 satellite imagery.

Site Discovery

- Crowd-sourcing projects get the public involved in site detection without requiring lots of training
- Potential to cover huge areas of land quickly (particularly with automation / machine learning)

New Hampshire Stone Wall Mapper



NH Stone Wall Mapper: <https://www.arcgis.com/apps/webappviewer/index.html?id=f4d57ec1a6b8414190ca0662456dffb0>

StoryMap about the project: <https://nhdes.maps.arcgis.com/apps/MapJournal/index.html?appid=8d0b2ac3e3ac40f4b12ab958402d7f36>

Site Discovery

- Remote sensing techniques can also detect archaeological features buried beneath the surface or otherwise obscured by modern features

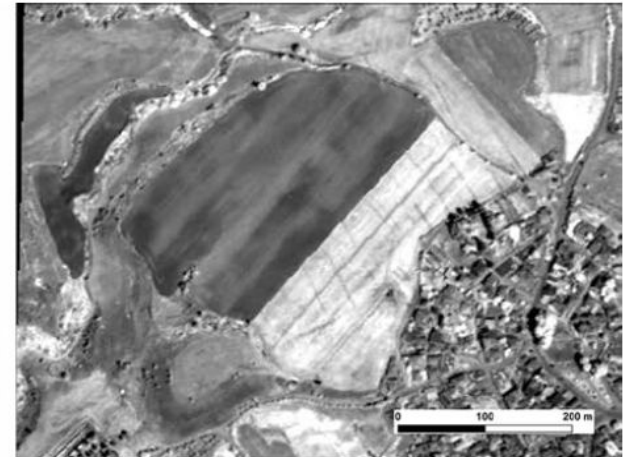
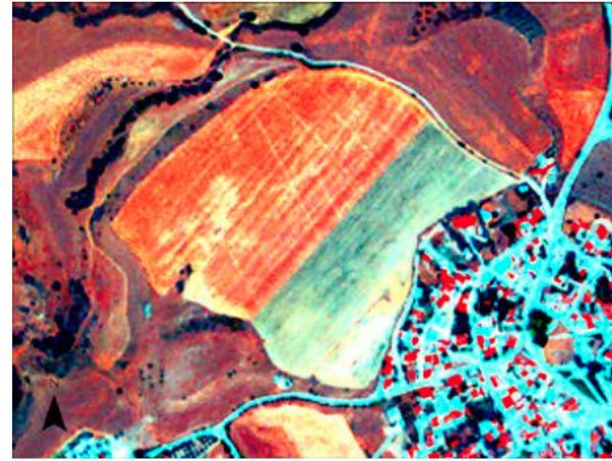
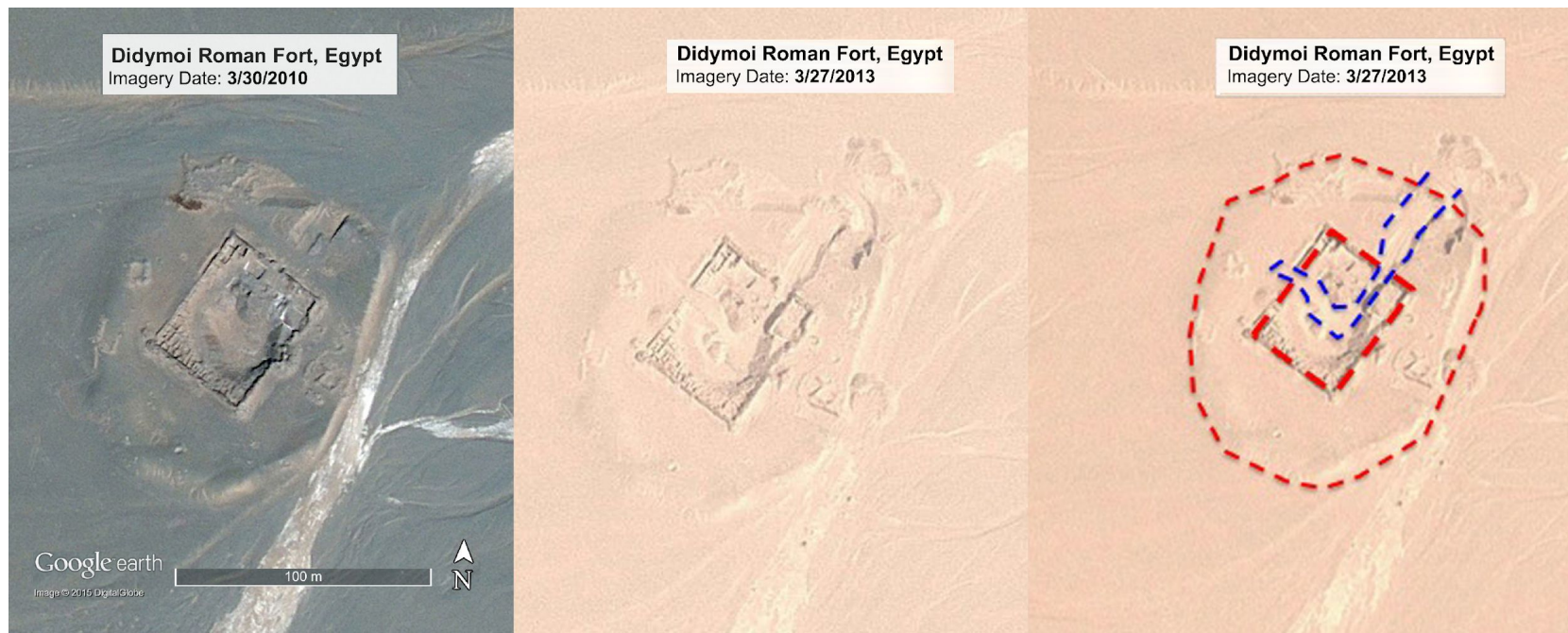


FIGURE 3. DECORRELATION STRETCH APPLIED TO QUICKBIRD, 15 JUNE 2009 (LEFT) AND MSR APPLIED TO GEOEYE-1, 4 MAY 2010 (RIGHT) SHOWING DIAGONAL SUBSURFACE STREETS AT PHERAI.

Donati, J. C. 2015. "Cities and Satellites: Discovering Ancient Urban Landscapes Through Remote Sensing Applications." In: *Best Practices of GeoInformatic Technologies for the Mapping of Archaeolandscapes*, edited by A. Sarris, 127–136. Oxford: Archaeopress.

Tracing change, monitoring damage

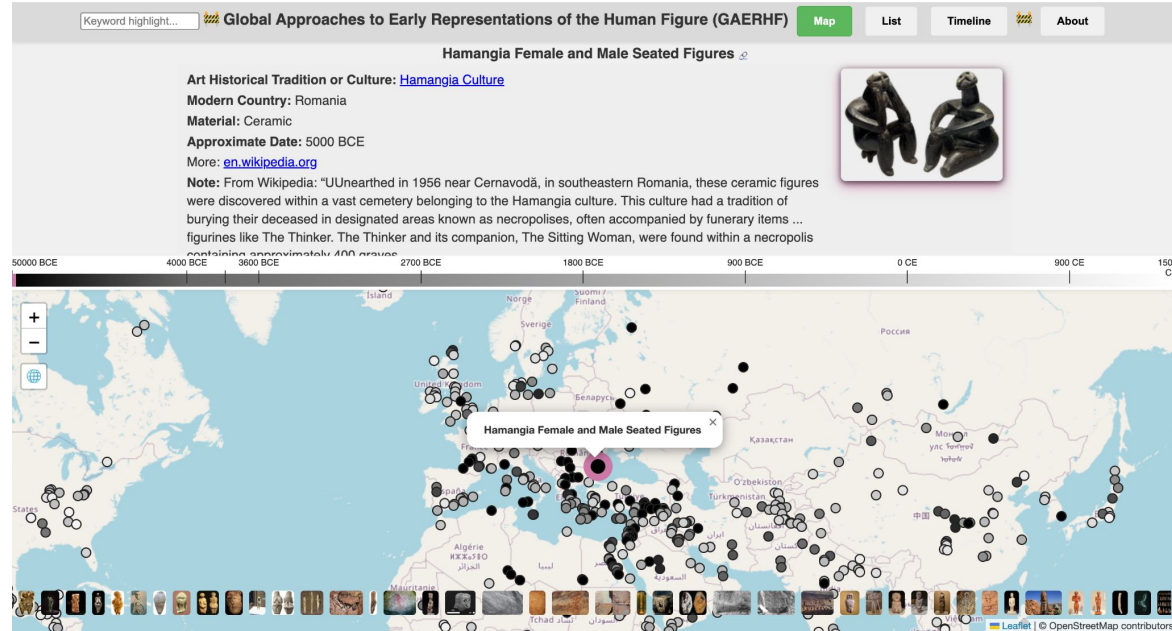


These images of the 1st century Roman Fort at Didymoi (eastern desert of Egypt) show the power of analyzing satellite images over time. On the left: an image of the fort in 2010; in the middle, the same view captured in 2013; on the right, identified areas of destruction on the site, most likely caused by bulldozers. Credits: EAMENA database: <https://database.eamena.org/>

Tracing objects and archives

- GIS and mapping can help track circulation of objects and retrace archival materials

<https://gaerhf.org/#hamangia-seated-figures>



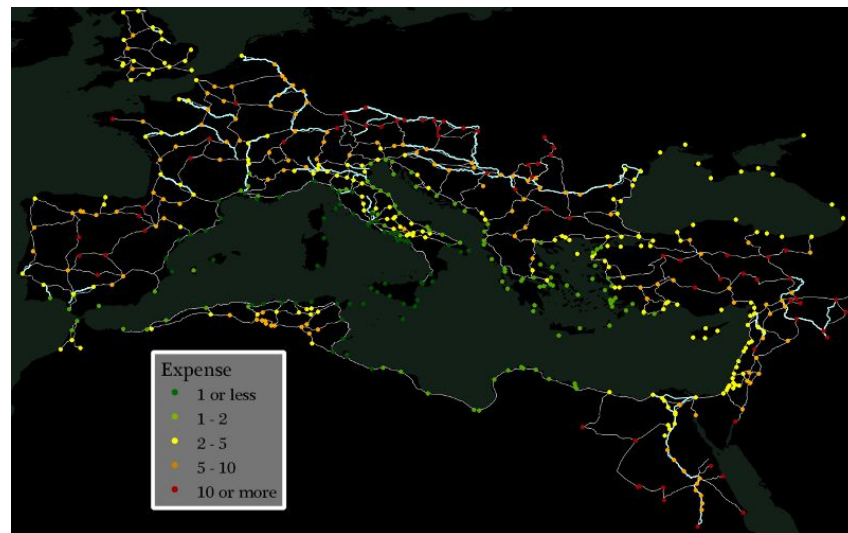
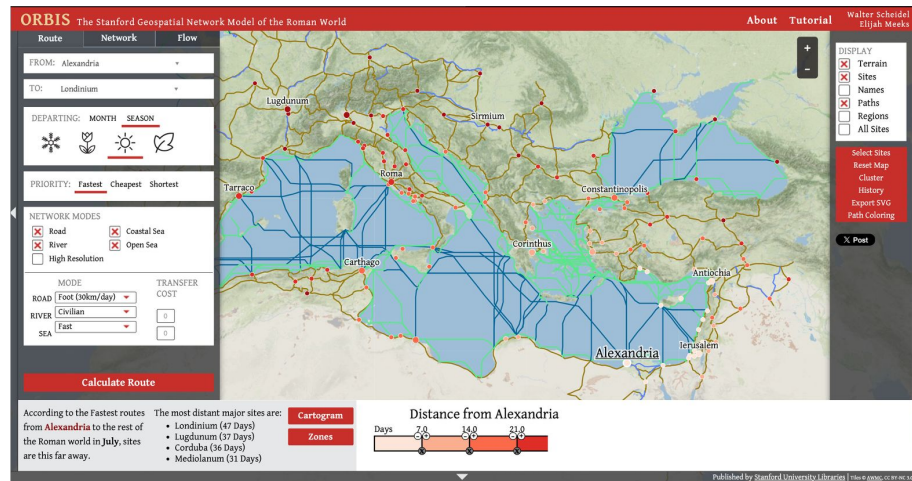
See Sunoikisis DCH 2026 Session 2: Collaborative Cultural Heritage Geodata

<https://github.com/SunoikisisDC/SunoikisisDC-2025-2026/wiki/2-Collaborative-Geodata>

Modelling large-scale mobility

ORBIS Stanford: a probabilistic model of travel in the 2nd-century Roman Empire

(<https://orbis.stanford.edu/>)

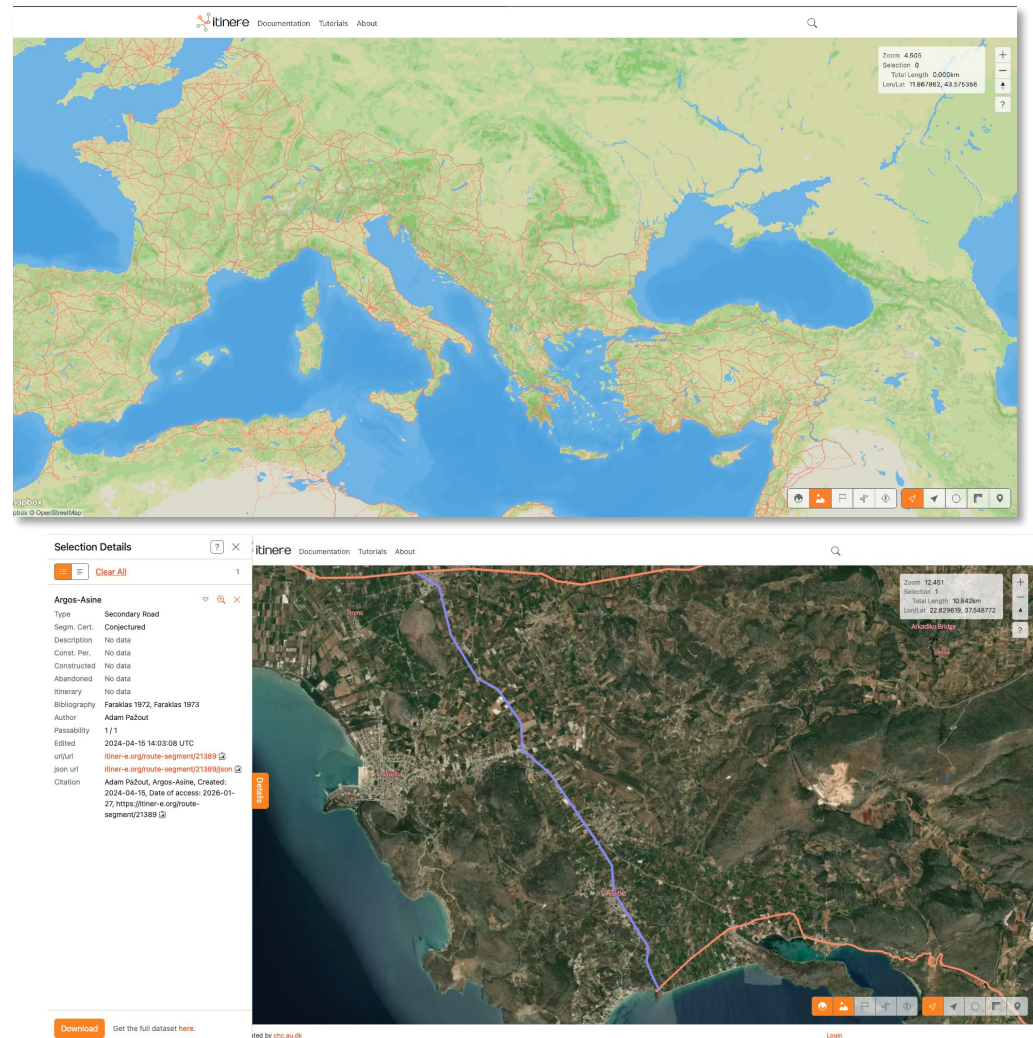


A cost distribution map for the Roman network, with points in different colours depending on calculated expense to reach the centre. Credits: W. Scheidel, Orbis Stanford.

Modelling mobility

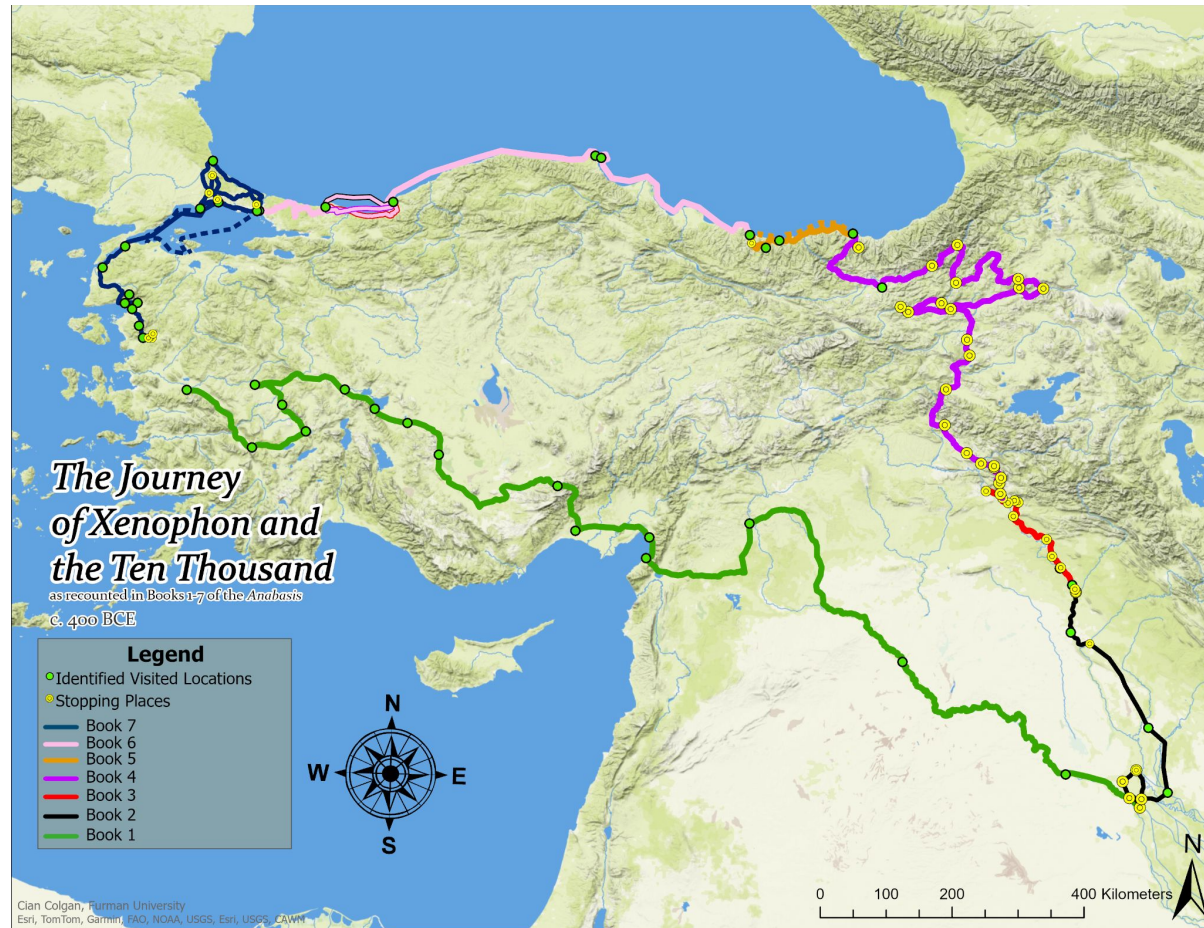
Itiner-e provides a connectivity-based model of the Roman Roads, with additional 3D layers and itinerary calculation

Brughmans, T., de Soto, P., Pažout, A. and Bjerregaard Vahlstrup, P. (2024) Itiner-e: the digital atlas of ancient roads.
<https://itiner-e.org>



Modelling complex travel narratives

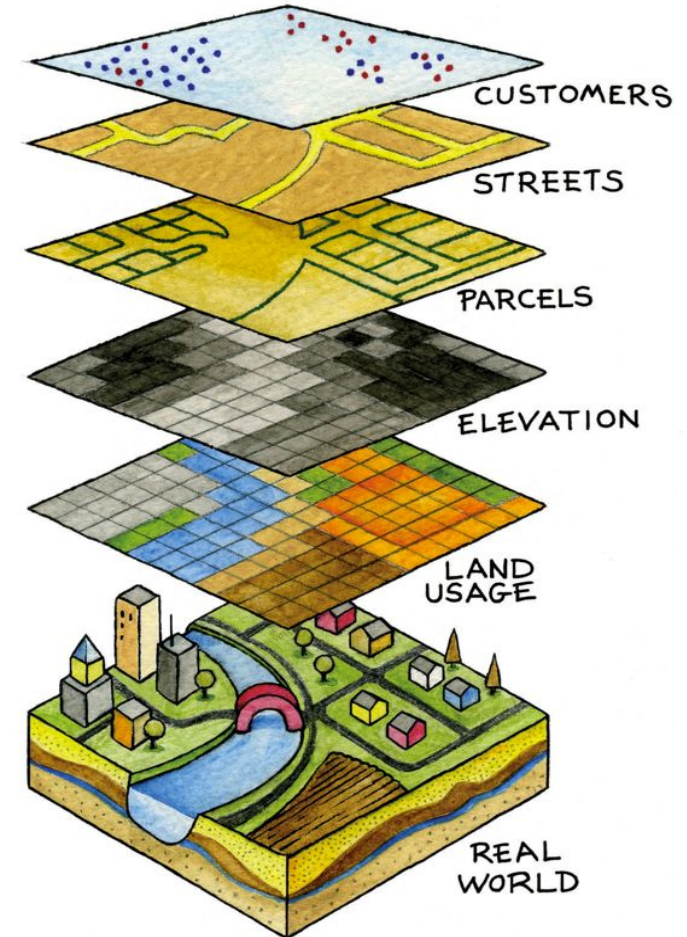
Using text annotation and GIS methods for spatial analysis, we can analyse spatially dense texts in new ways



The Basic Concepts of GIS

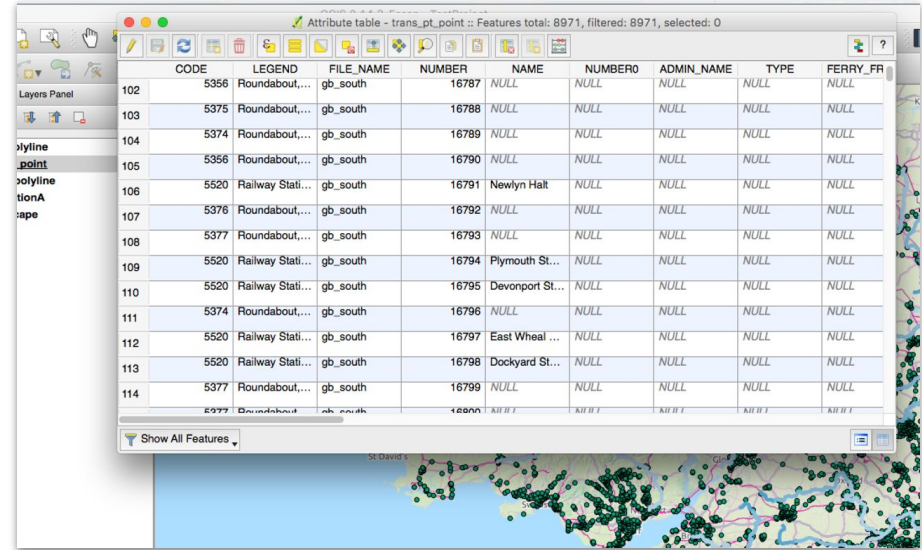
GIS as Layers

- GIS data are essentially an abstraction of the real world
- Each *kind of thing* in the real world is usually represented by its own GIS data layer
- Layers at the top of the stack are rendered above lower layers



Most GIS data is made up of:

1. Location(s) in space, e.g. XY coordinate pair
2. Some kind of additional data, e.g. a table containing attribute information (id, name, size, date) or a value (elevation, RGB color)



Attribute table - trans_pt_point :: Features total: 8971, filtered: 8971, selected: 0

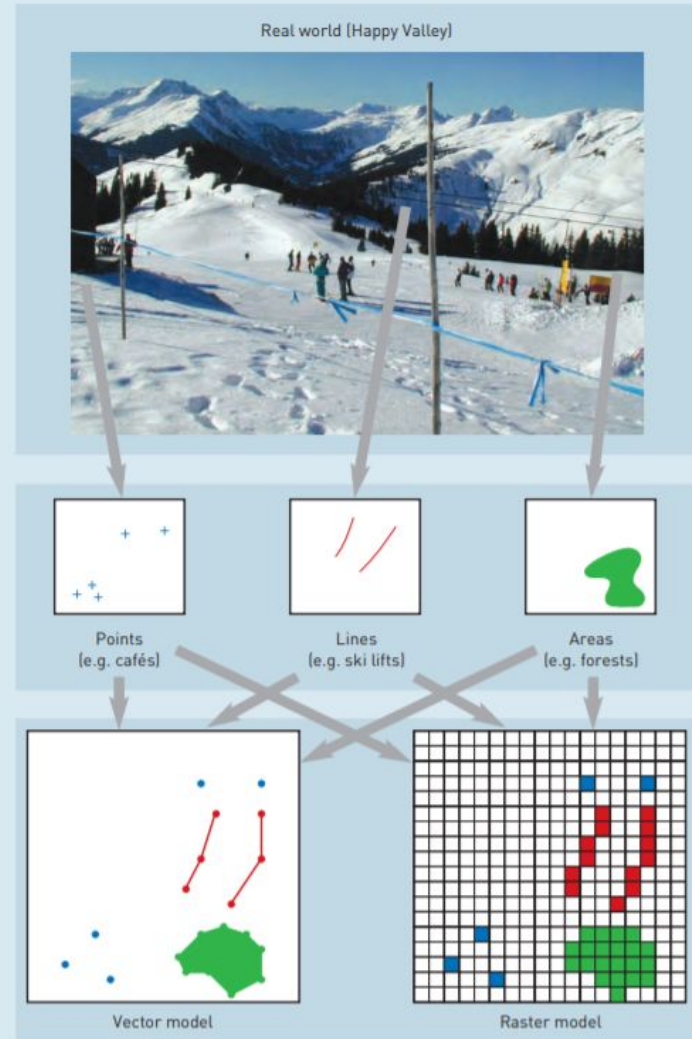
	CODE	LEGEND	FILE_NAME	NUMBER	NAME	NUMBER0	ADMIN_NAME	TYPE	FERRY_FR
102	5356	Roundabout...	gb_south	16787	NULL	NULL	NULL	NULL	NULL
103	5375	Roundabout...	gb_south	16788	NULL	NULL	NULL	NULL	NULL
104	5374	Roundabout...	gb_south	16789	NULL	NULL	NULL	NULL	NULL
105	5356	Roundabout...	gb_south	16790	NULL	NULL	NULL	NULL	NULL
106	5520	Railway Stati...	gb_south	16791	Newlyn Halt	NULL	NULL	NULL	NULL
107	5376	Roundabout...	gb_south	16792	NULL	NULL	NULL	NULL	NULL
108	5377	Roundabout...	gb_south	16793	NULL	NULL	NULL	NULL	NULL
109	5520	Railway Stati...	gb_south	16794	Plymouth St...	NULL	NULL	NULL	NULL
110	5520	Railway Stati...	gb_south	16795	Devonport St...	NULL	NULL	NULL	NULL
111	5374	Roundabout...	gb_south	16796	NULL	NULL	NULL	NULL	NULL
112	5520	Railway Stati...	gb_south	16797	East Wheel ...	NULL	NULL	NULL	NULL
113	5520	Railway Stati...	gb_south	16798	Dockyard St...	NULL	NULL	NULL	NULL
114	5377	Roundabout...	gb_south	16799	NULL	NULL	NULL	NULL	NULL
115	5377	Roundabout...	gb_south	16800	NULL	NULL	NULL	NULL	NULL

Show All Features

Vectors vs. Rasters

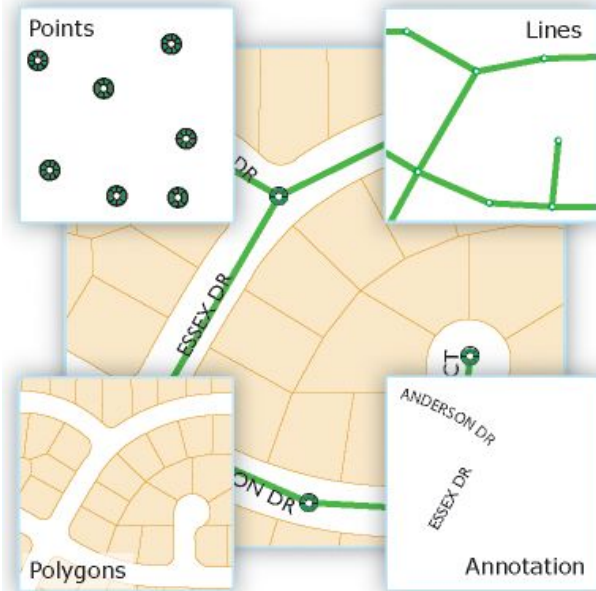
- Vector and raster data structures store, represent and manipulate spatial information in different ways.
- They can both be used to represent the real world!

Figure 1.17. Heywood, Ian, Sarah Cornelius & Steve Carver. 2006. An introduction to geographical information systems, 3rd edn. Harlow, UK: Pearson Prentice Hall.



Vectors

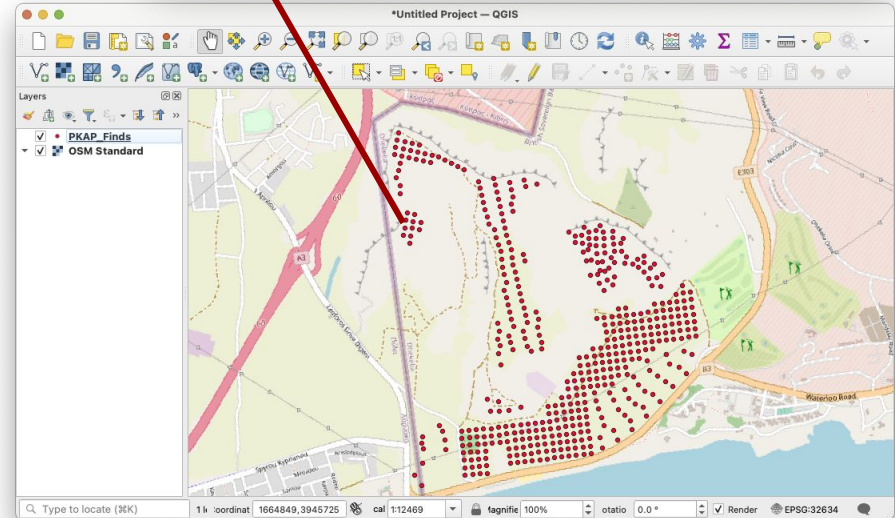
- Represent the real world as **points**, **lines**, or **polygons**
- Information about each feature is stored in a linked **attribute table**



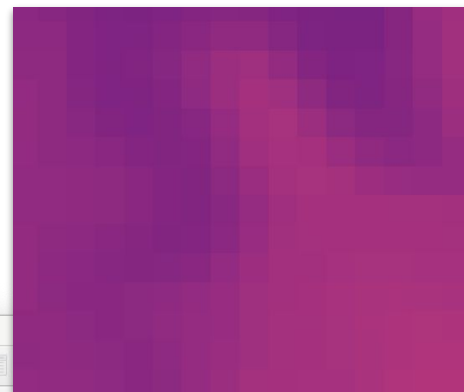
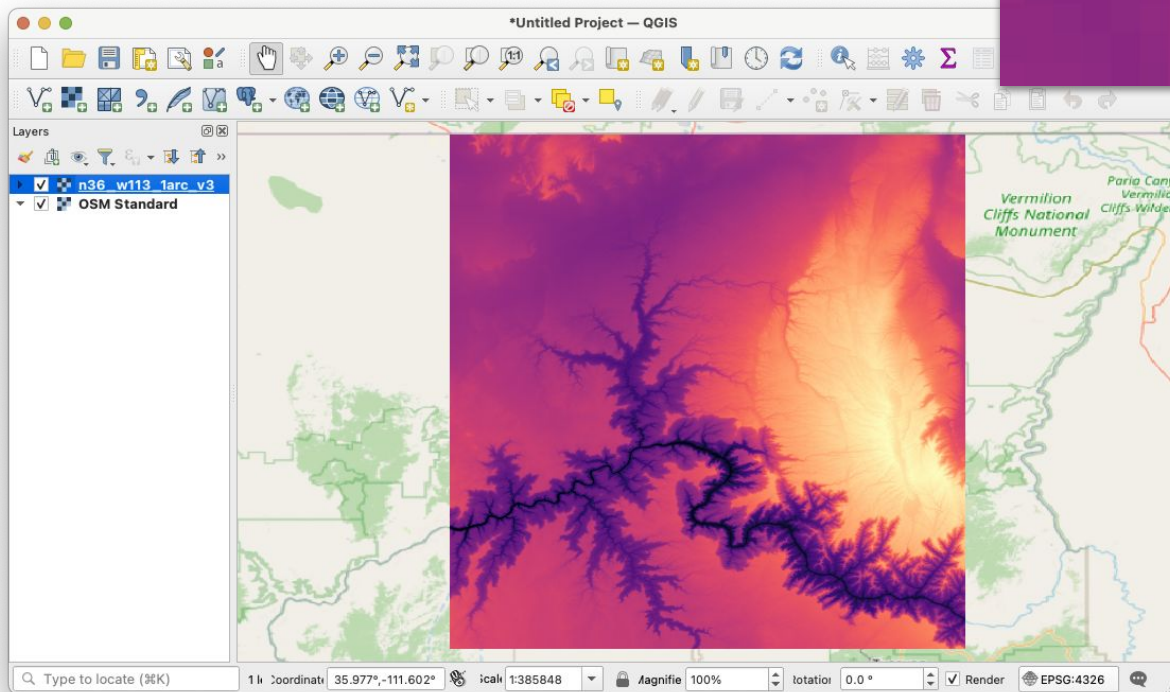
PKAP_Finds — Features Total: 465, Filtered: 465, Selected: 0

	Unit	Latitude	Longitude	Quantity	Bone_Shell	Glass
1	Unit 1	34.981810	33.706001	44.000000	0	2.000000
2	Unit 2	34.981807	33.706439	64.000000	0	2.000000
3	Unit 3	34.981805	33.706877	59.000000	0	1.000000
4	Unit 4	34.981802	33.707316	57.000000	0	1.000000
5	Unit 5	34.981800	33.707754	65.000000	0	0
6	Unit 6	34.982163	33.707319	83.000000	0	0
7	Unit 7	34.982166	33.706880	59.000000	0	0

Show All Features

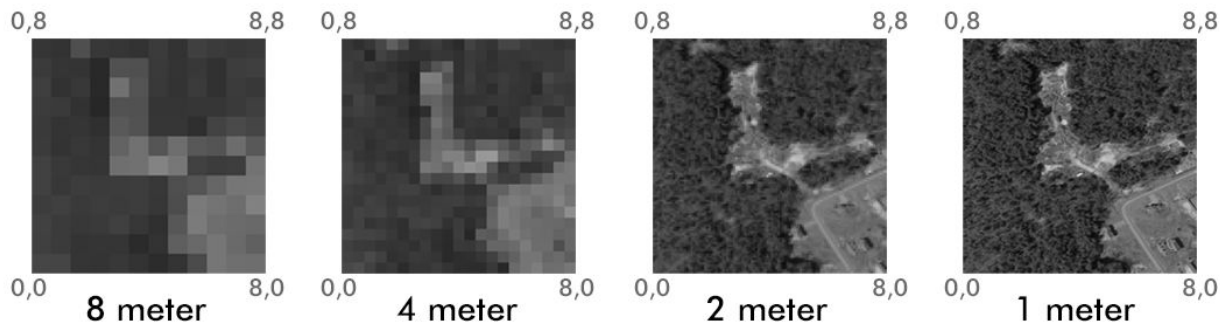
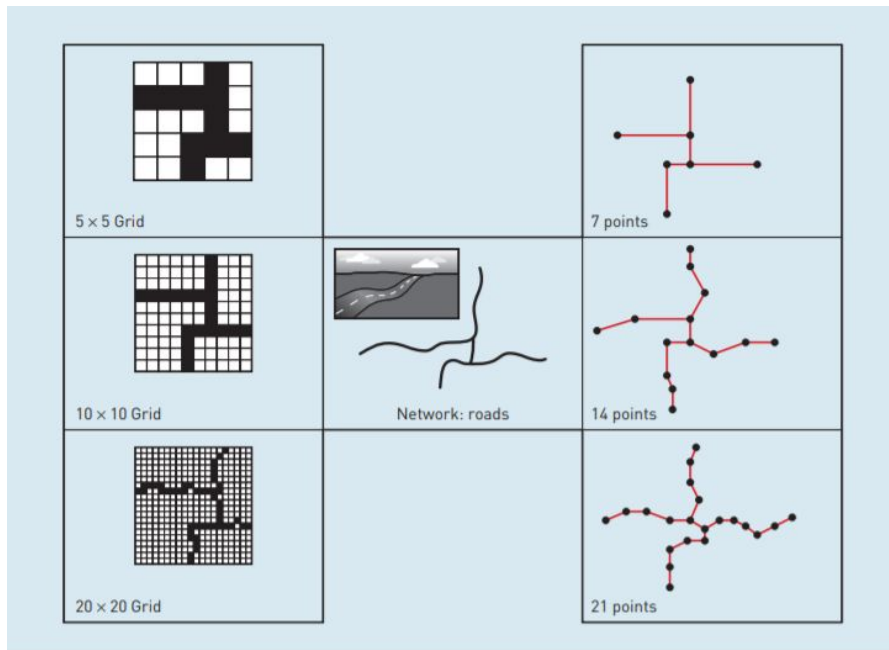


Rasters



- Represent the real world as a grid of **cells** (or pixels)
- Each cell has a **value** representing a measurement (like elevation) or an RGB color value

Vectors and
rasters both
come in different
resolutions

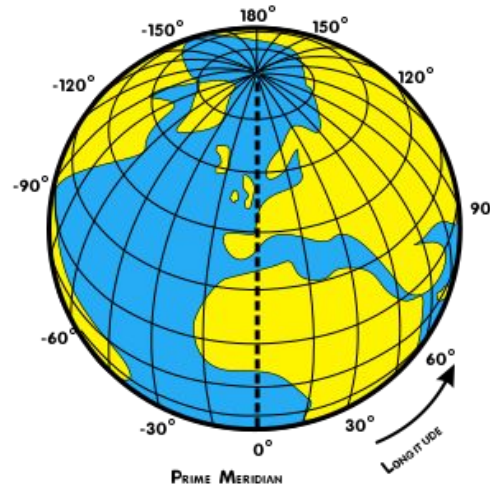
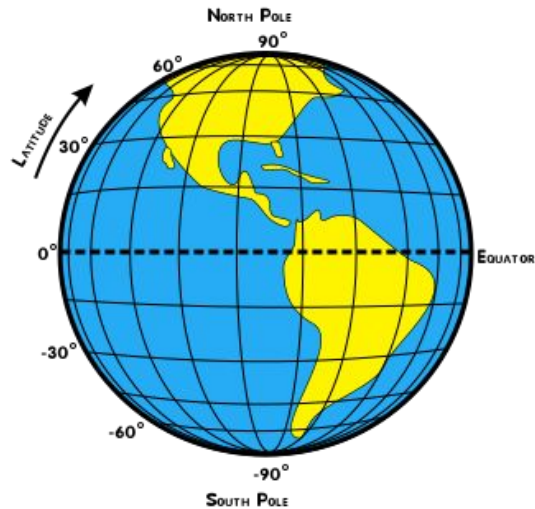


Coordinate Reference Systems (CRS's)

If I told you it was 40 degrees today, what would you imagine?

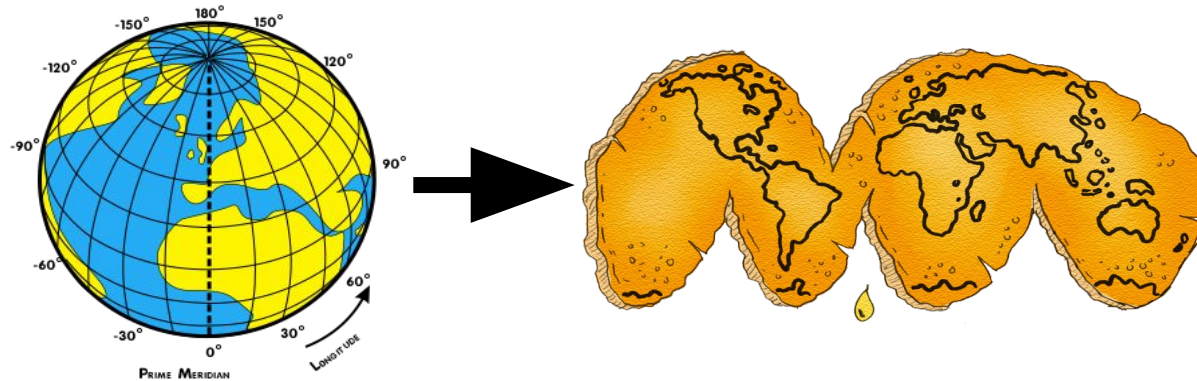


In **geographic** coordinate systems, location is represented by the intersection of latitude and longitude lines.



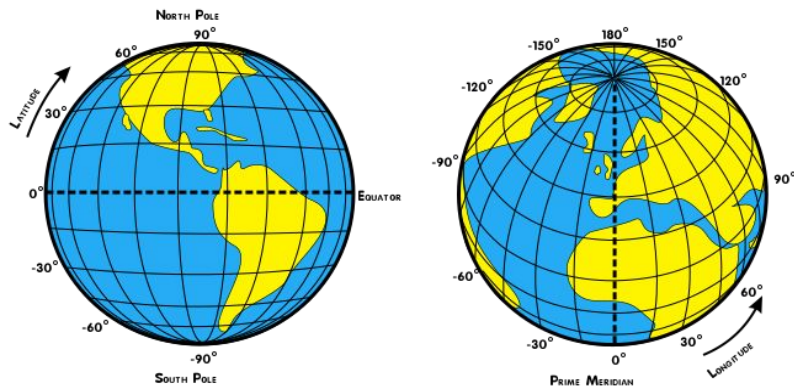
Amherst, MA =
42.3732° (N)
-72.5199° (W)
WGS84

Projected coordinate systems use mathematical technique to show features from a spherical surface as if they were on a flat surface. Location is represented as a distance (often in meters) from a 0 point.



Amherst, MA =
704197 (X)
4694193 (Y)
WGS84 UTM Zone 18N

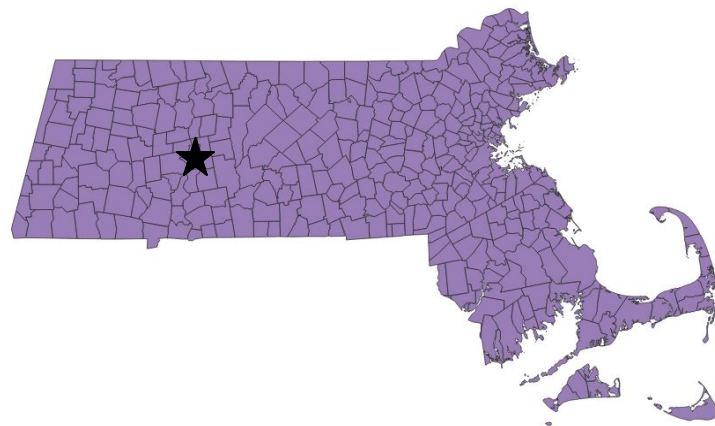
Coordinates can be converted from a Geographic CRS to a Projected CRS:



Amherst, MA

WGS84

42.3732° N, 72.5199° W

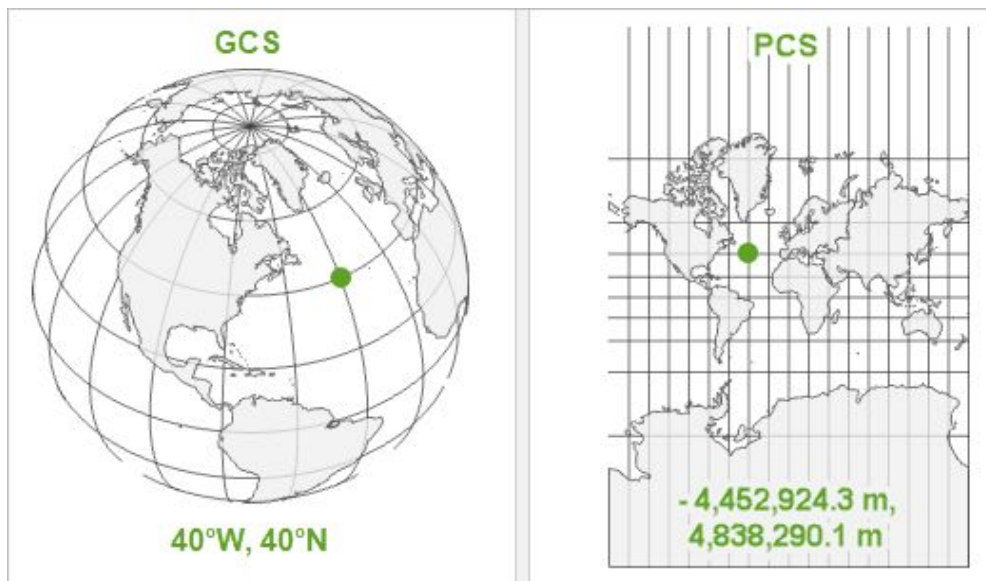


Amherst, MA

WGS84 UTM Zone 18N

704197, 4694193

For 3D surfaces!



**Geographic
CRS's**

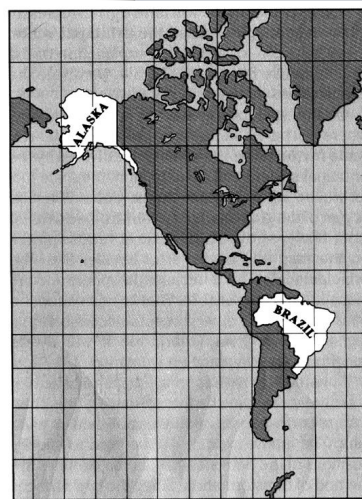
**Projected
CRS's**

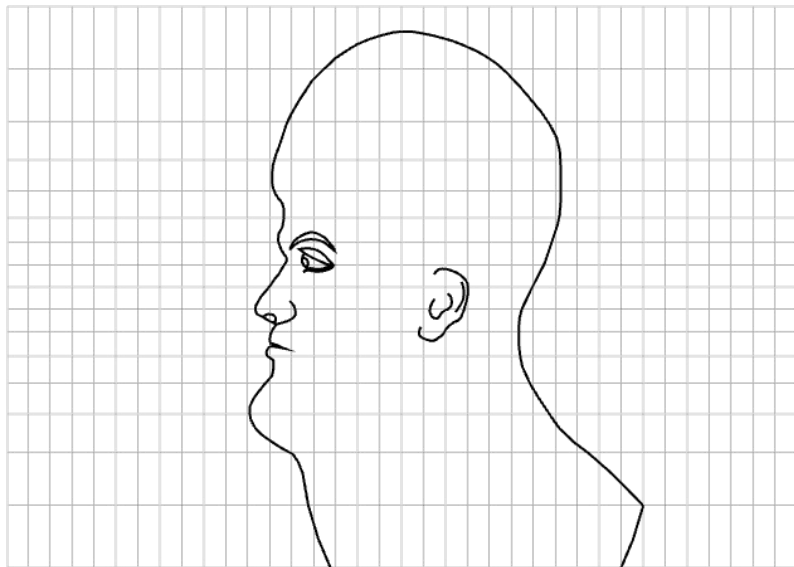
**For flat
surfaces!**

Projection = distortion

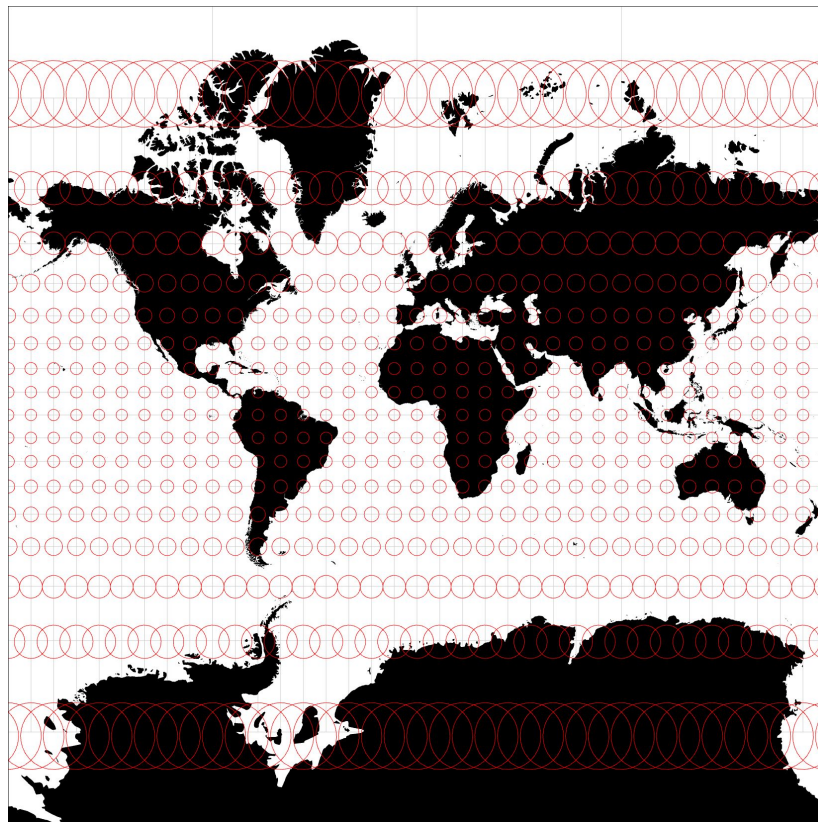
Projections always involve some kind of distortion in **shape**, **distance**, or **area**. Each projection tries to minimize one or more of these types of distortion.

The **Mercator** projection preserves **shape**, but distorts **area** and **distance**





Created by [Andy Woodruff](#) from examples by [Jason Davies](#) and [Mike Bostock](#). Modified by Garrett Dash Nelson for *Bending Lines*.

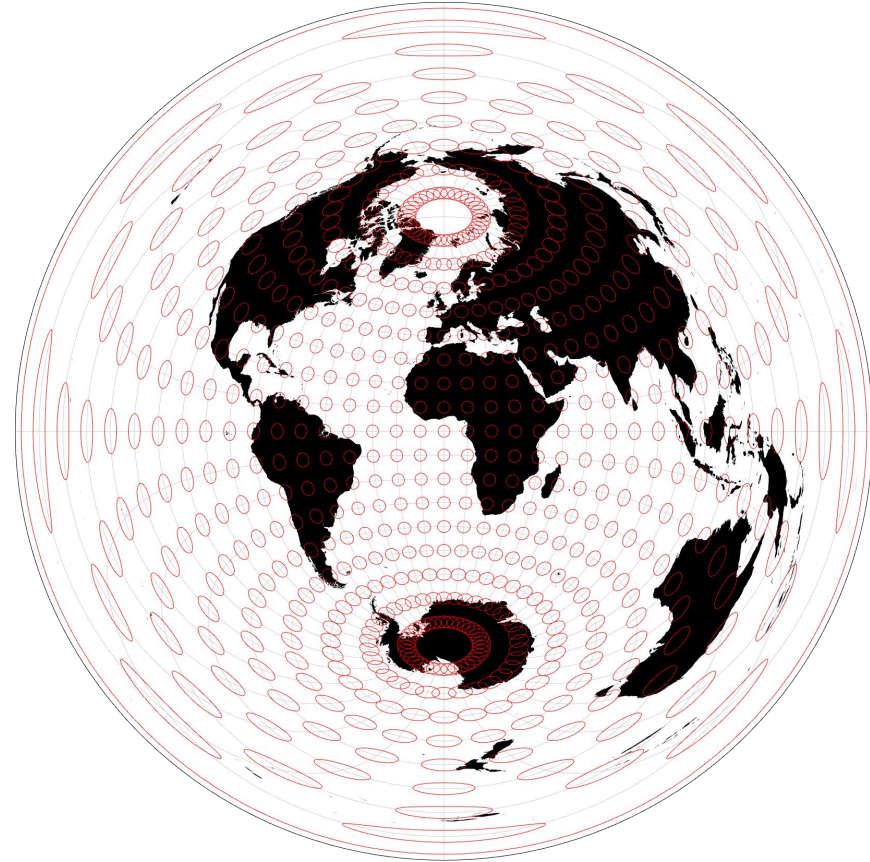


Tissot's Indicatrix by [Mike Bostock](#) on [Observable](#).

The **Azimuthal Equidistant** projection preserves **distance** from the center of the map but distorts **shape** and **area**



United Nations logo

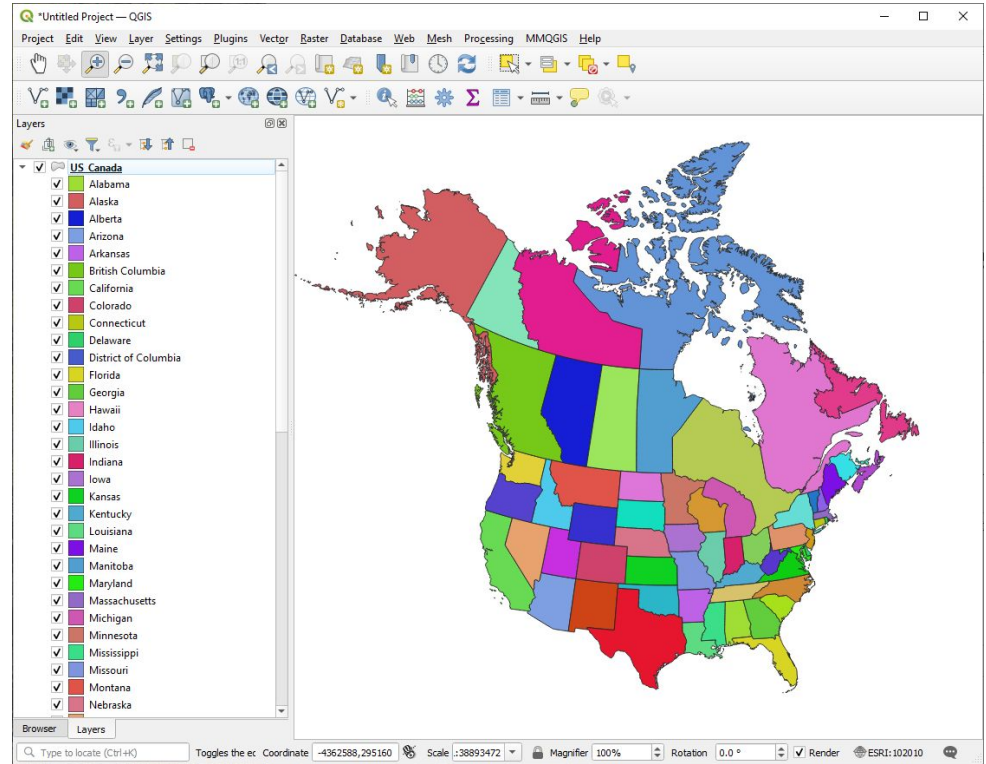


There are two CRS's for you to keep in mind:

- Each **data layer** has its own CRS. This geometry information is encoded into the file itself and cannot be changed.
 - Open a layer's properties and navigate to the "Information" tab, then look for "Coordinate Reference System (CRS)."
- The **QGIS project** has its own CRS. Any data layers in a different system will be "projected on the fly" into this CRS.
 - In the bottom right corner, click the "EPSG:####" button, or open Project > Properties and navigate to the "CRS" tab.

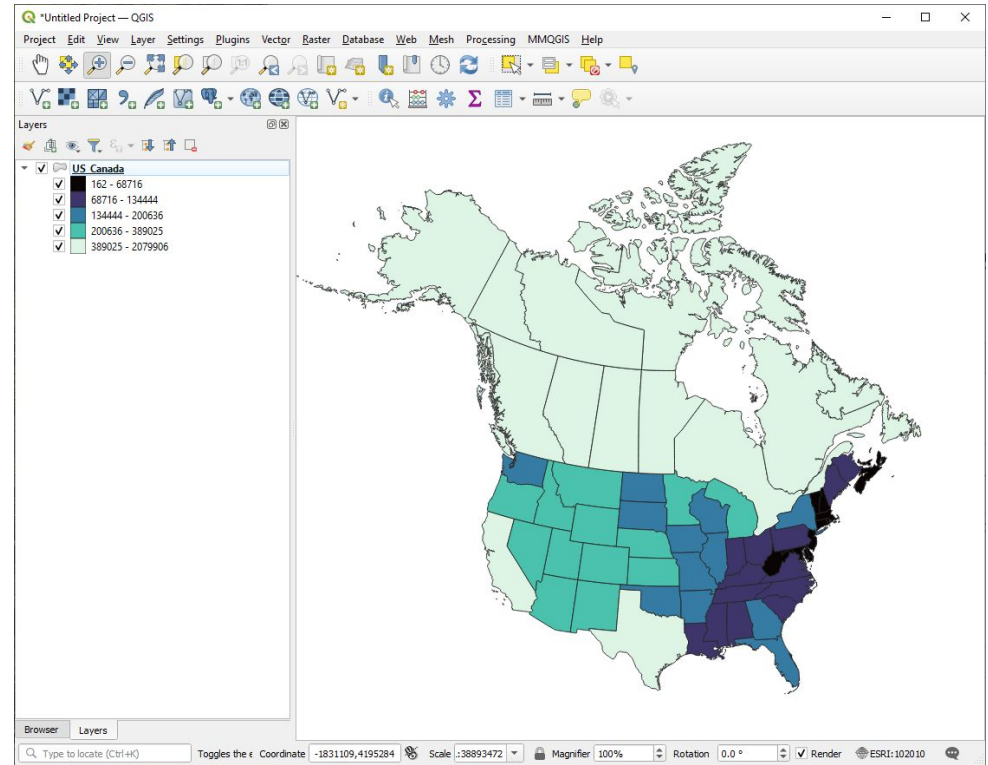
Symbology

We can use the values stored in a vector's attribute table or a raster's cells to change the way the data looks, including the **size**, **color**, **transparency**, or **labels**.



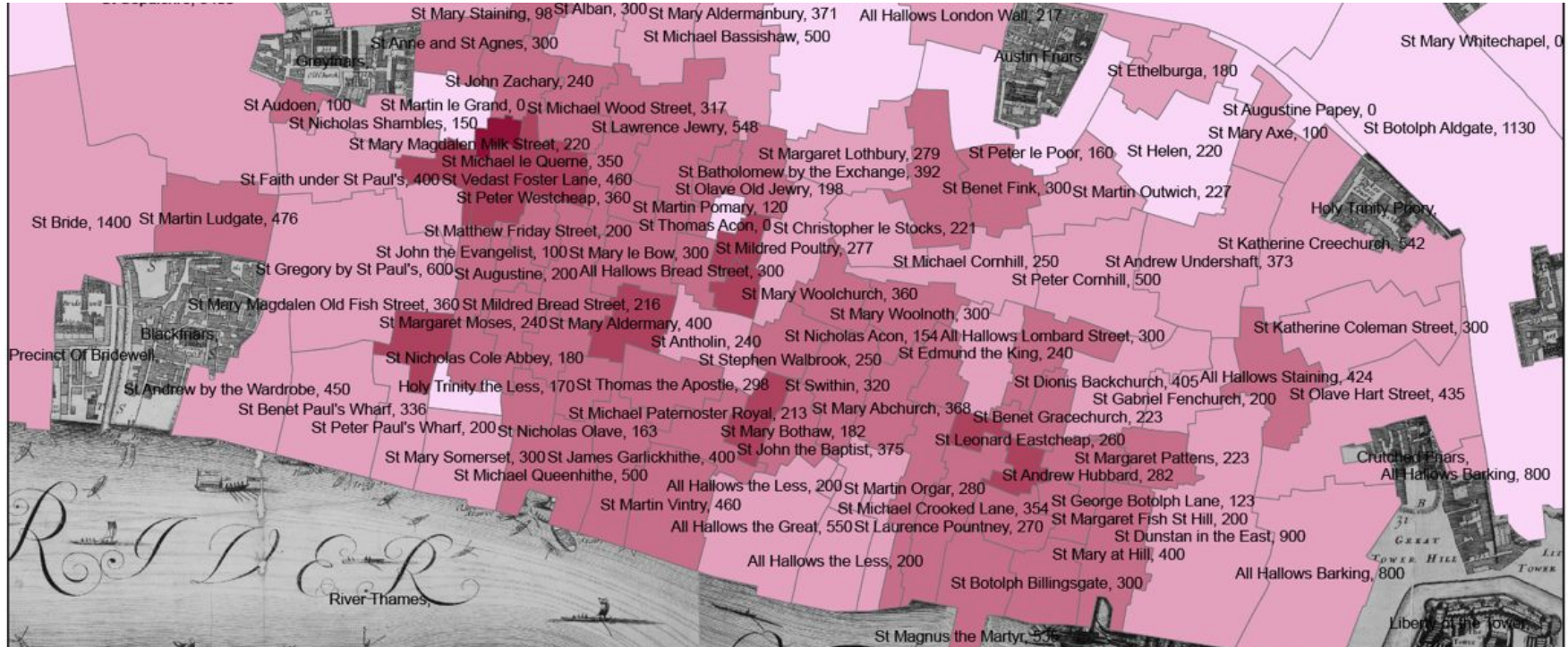
Using a **categorized** (or unique) color scheme based on state or province name

We can use the values stored in a vector's attribute table or a raster's cells to change the way the data looks, including the **size**, **color**, **transparency**, or **labels**.

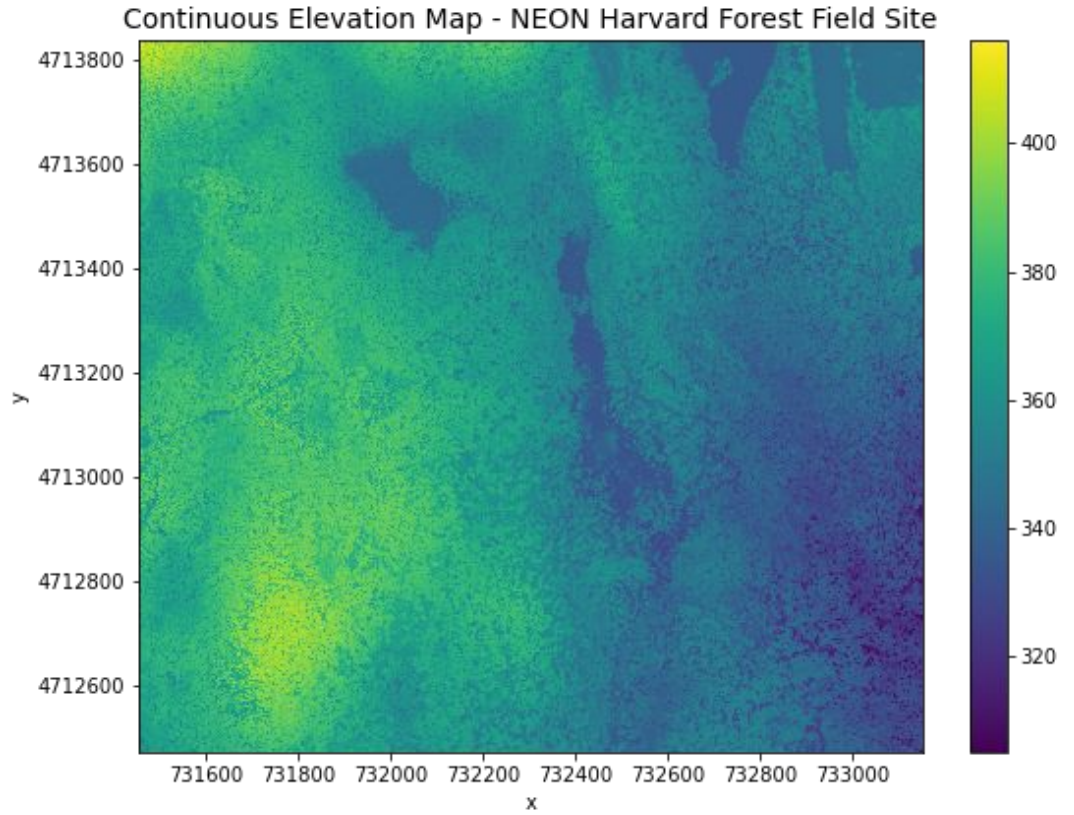


Using a **graduated** color scheme to show area in sq. km

Using attributes to apply **labels** and shade the polygons based on **population density**



Shading a raster based on the **elevation** values within each cell



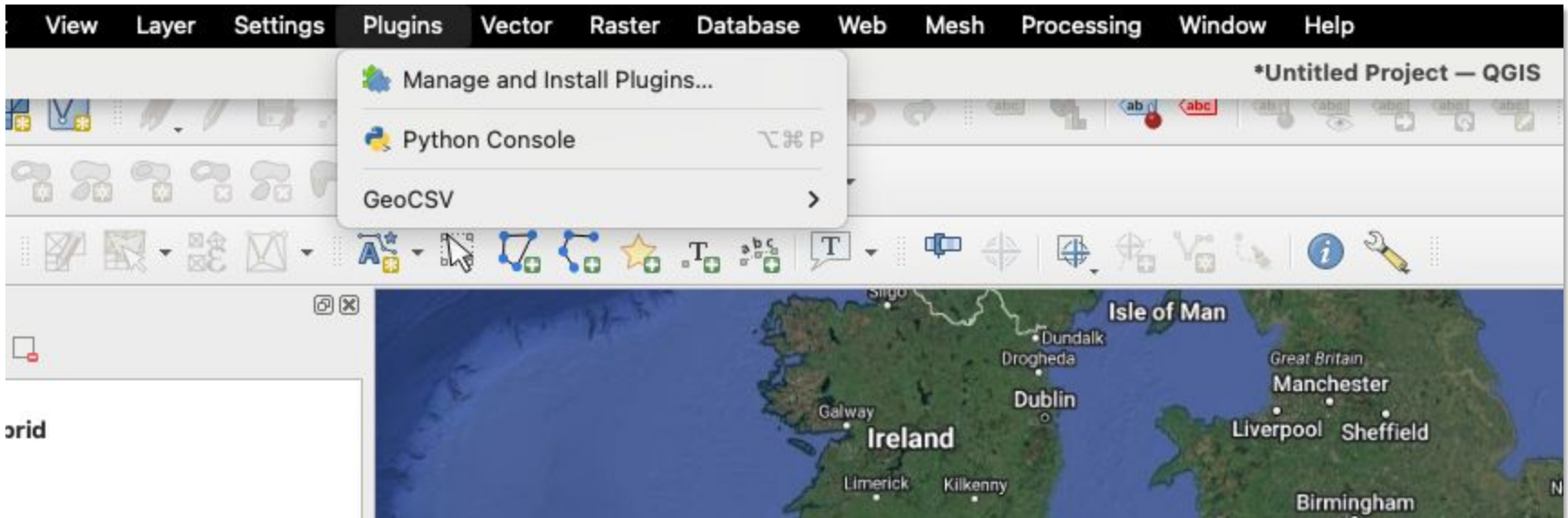
Demo: Getting started with QGIS

INSTALLING AND OPENING QGIS

1. Go to <https://qgis.org> and click on “Download”.
2. Download and install the most recent Long-Term Version or Regular Version.
3. Open it on your computer (you may need a workaround if you’re opening this on a Mac for the first time - See GitHub instructions).
4. Create a New Empty Project.

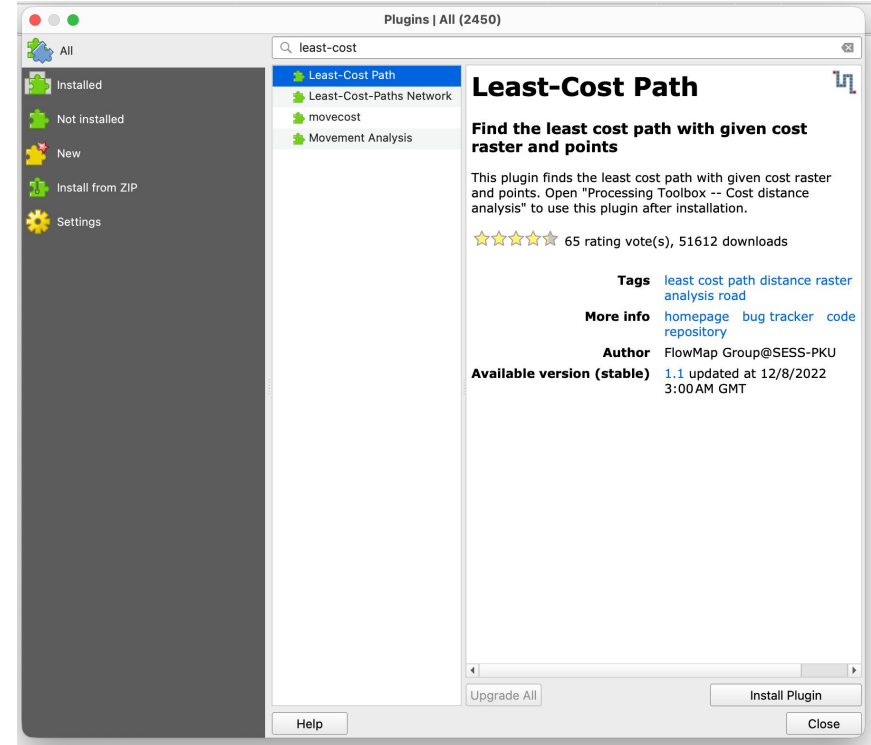
INSTALL THE PLUGINS

1. Menu bar > Plugins > Manage and Install



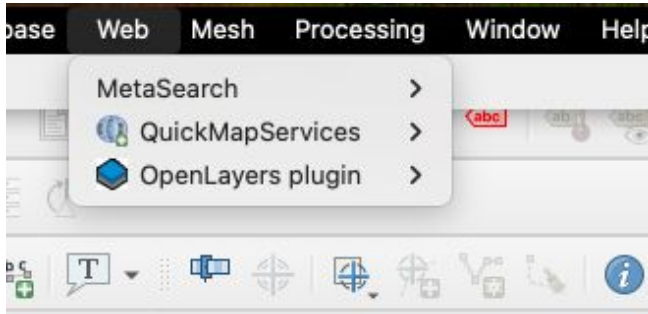
INSTALL THE PLUGINS

1. Open QGIS > New Empty Project
2. Menu bar > Plugins > Manage and Install
3. Search for “QuickMapServices” and install
4. Search for “Least-Cost Path” and install



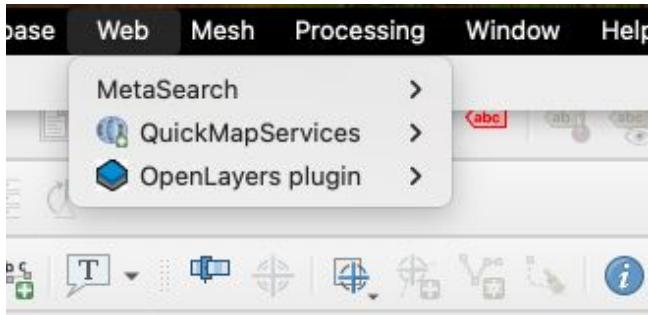
ACCESS THE QMS PLUGIN

1. Menu bar > Web > QuickMapServices

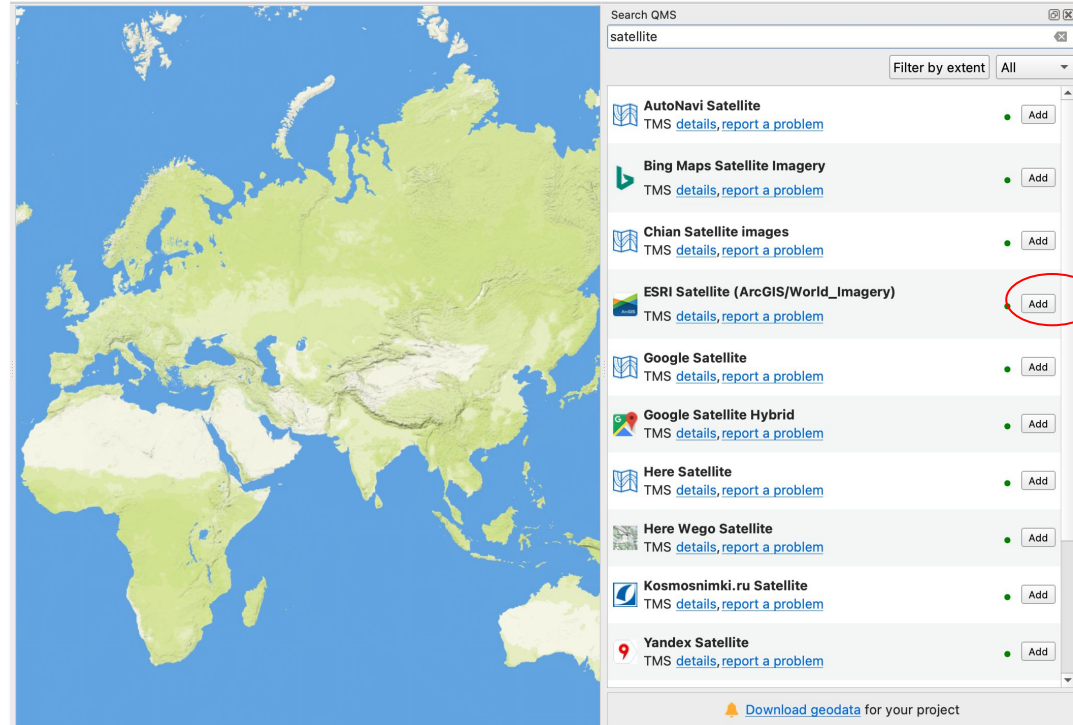
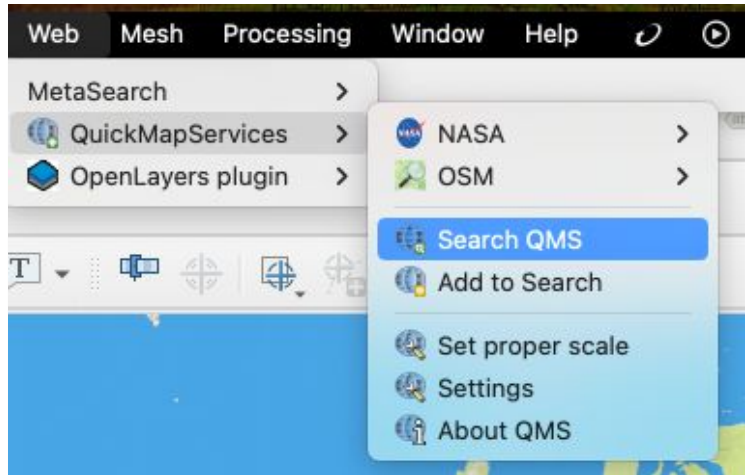


ACCESS THE QMS PLUGIN

1. Menu bar > Web > QuickMapServices

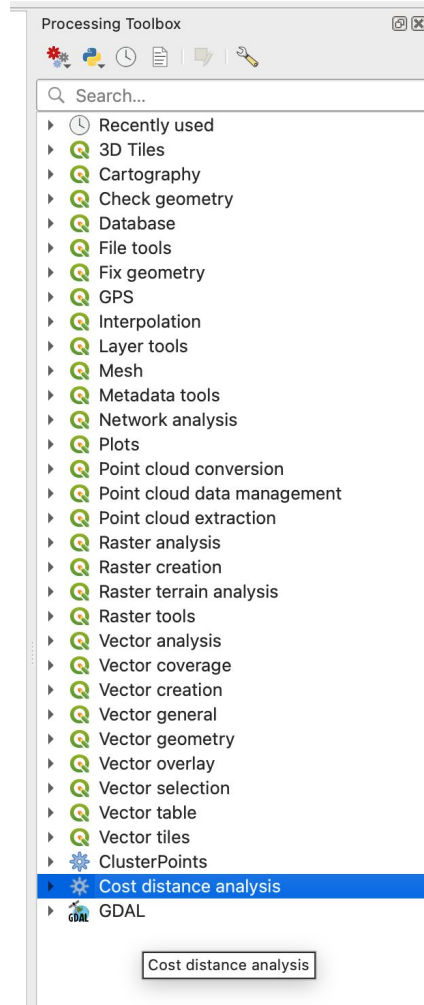
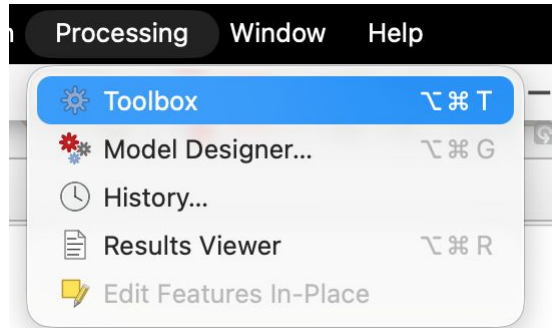


ACCESS THE QMS PLUGIN



ACCESS THE PROCESSING TOOLBOX AND THE LEAST-COST PLUGIN

1. Menu > Processing > Toolbox > scroll down to “Cost distance analysis”

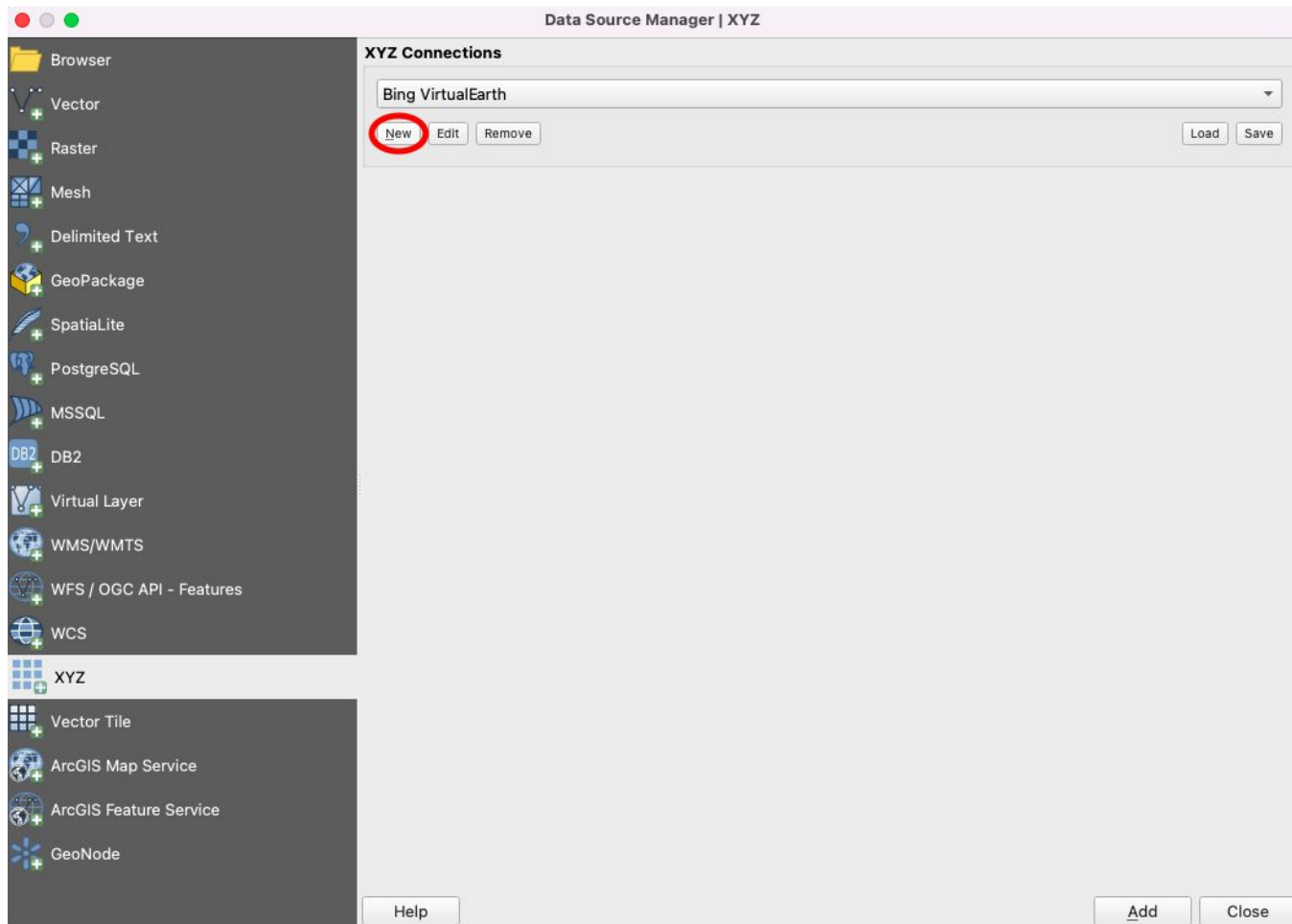


USING THE AWMC MAP TILE

Step-by-step Instructions:
<https://cawm.lib.uiowa.edu/xyz.html>

1. Menu > Layer > Add Layer
2. Add XYZ Layer
3. New Connection
4. Copy the URL:
<http://cawm.lib.uiowa.edu/tiles/{z}/{x}/{y}.png>
5. Then “Add”





URL:

<http://cawm.lib.uiowa.edu/tiles/{z}/{x}/{y}.png>

XYZ Connection

Connection Details

Name

URL

Authentication

Configurations Basic

Choose or create an authentication configuration

No Authentication

Configurations store encrypted credentials in the QGIS authentication database.

☒ Min.Zoom Level

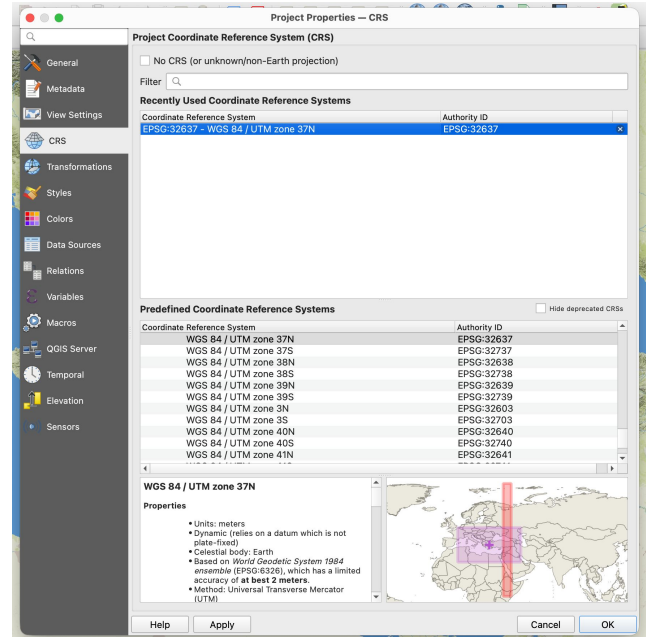
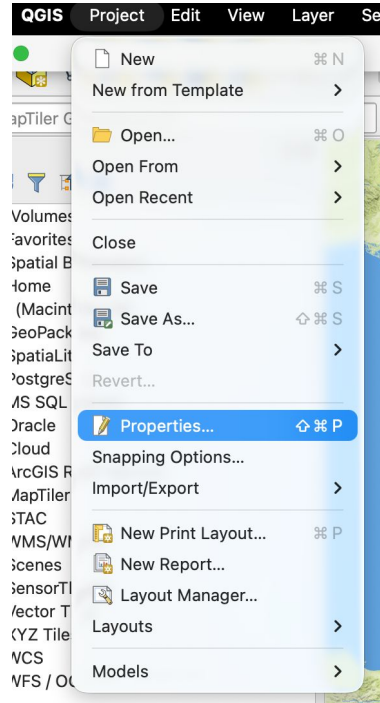
☒ Max.Zoom Level

Referer

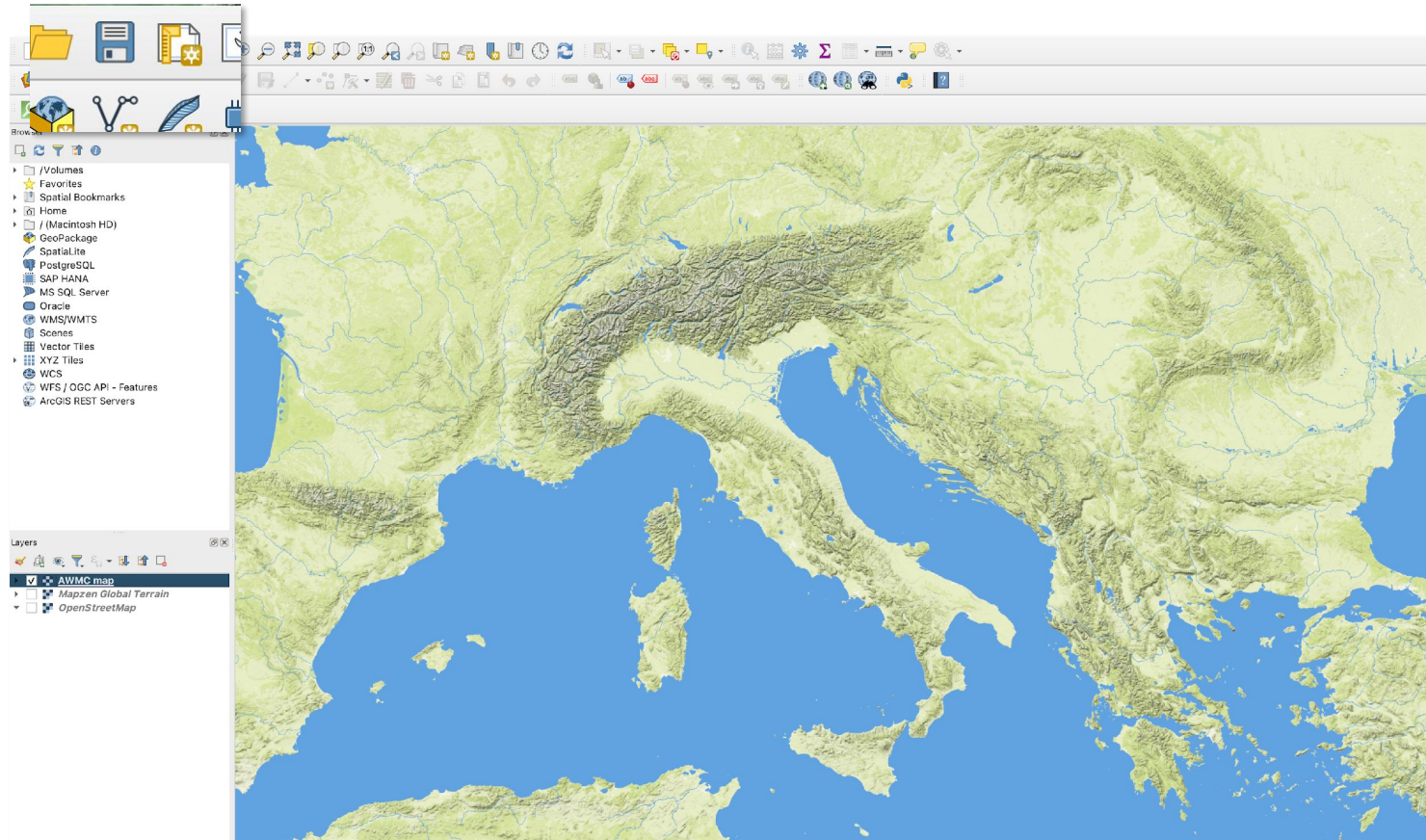
Tile Resolution

Cancel

Set the Coordinate Reference System



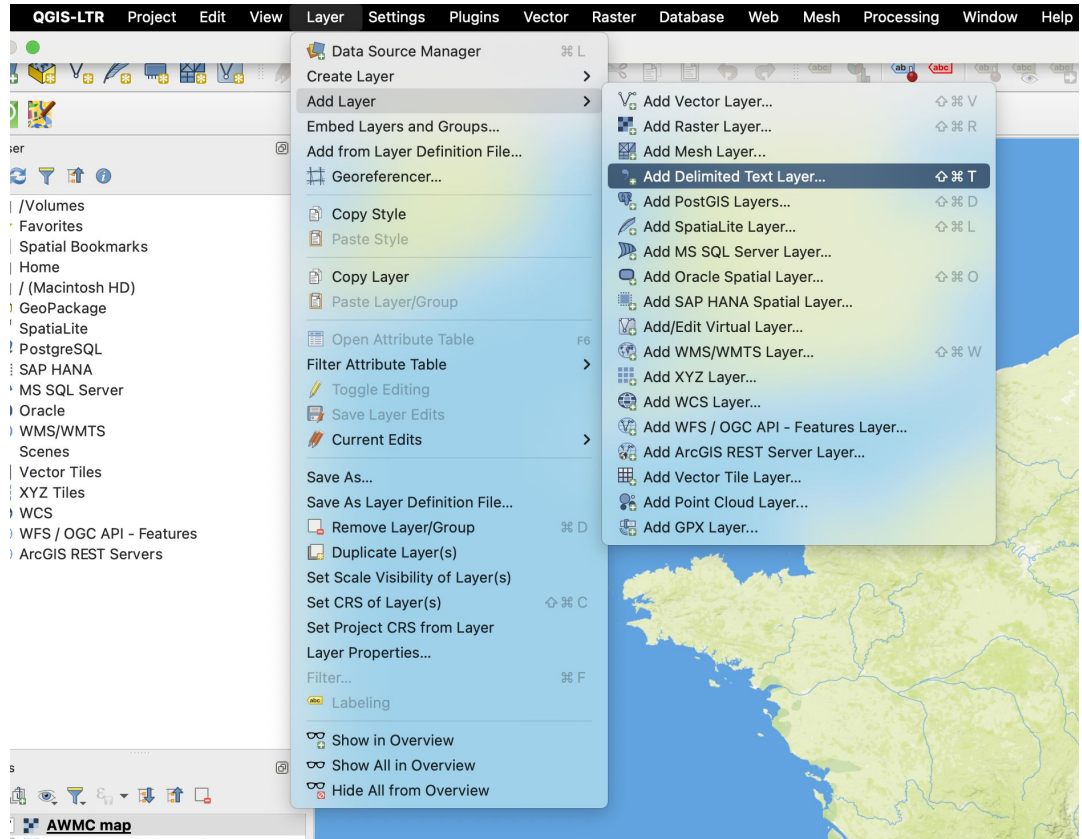
SAVE YOUR PROJECT



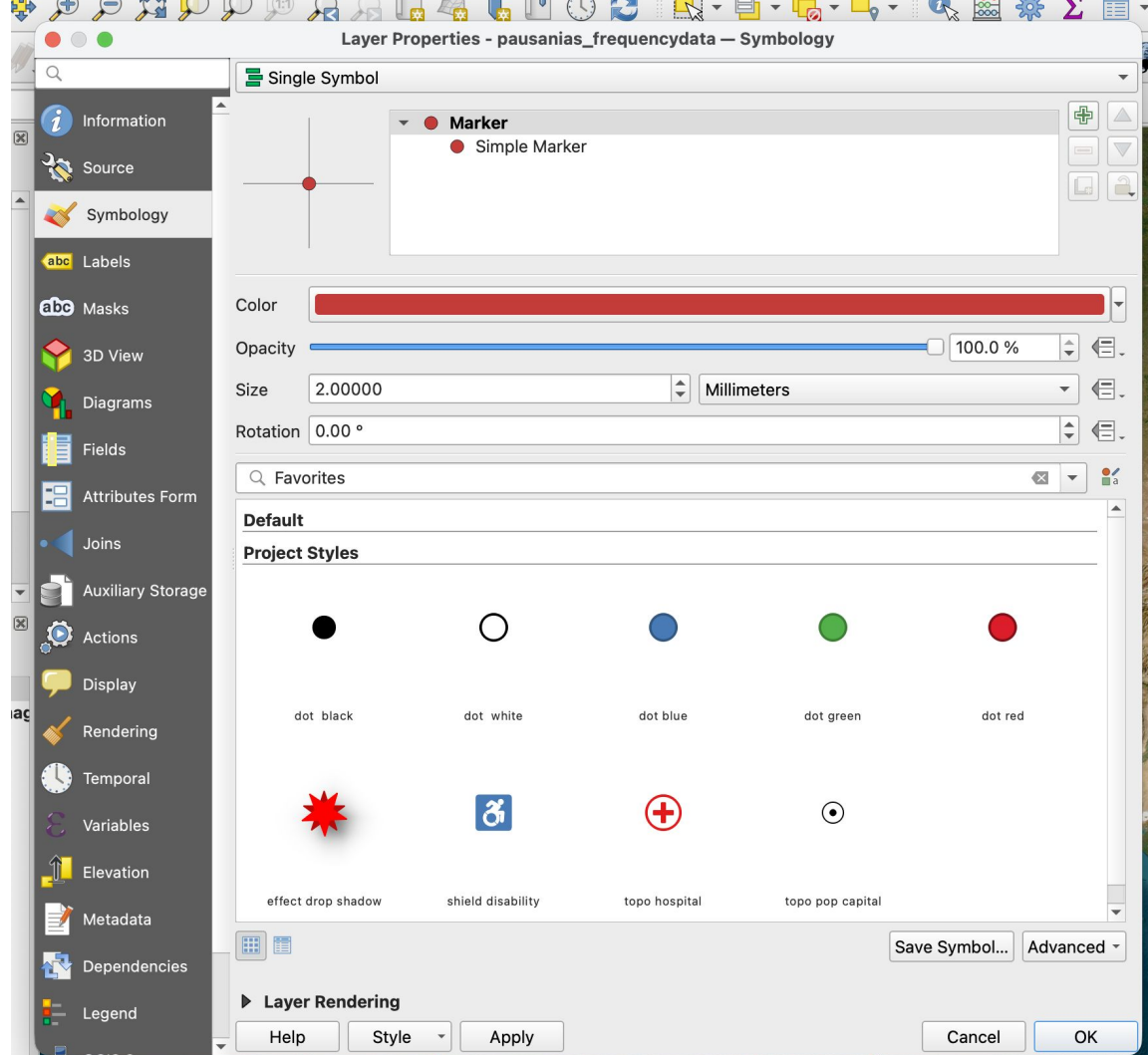
UPLOADING AND STYLING DATA



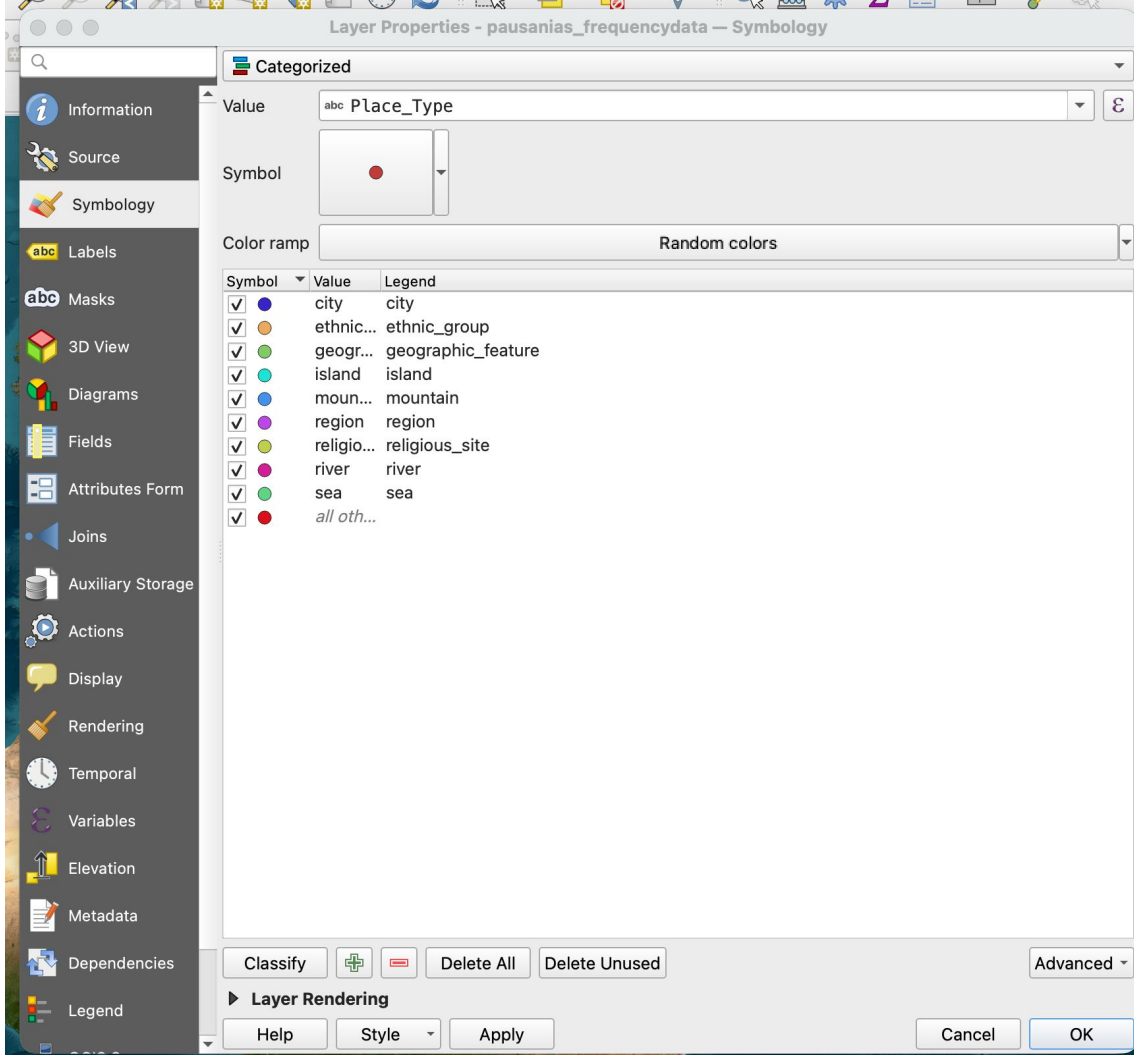
1. Shapefiles (.shp) -> simply drag and drop!
2. Tables -> Add delimited text layer
3. JSON files -> add Vector layer
4. Image files -> Add Raster Layer



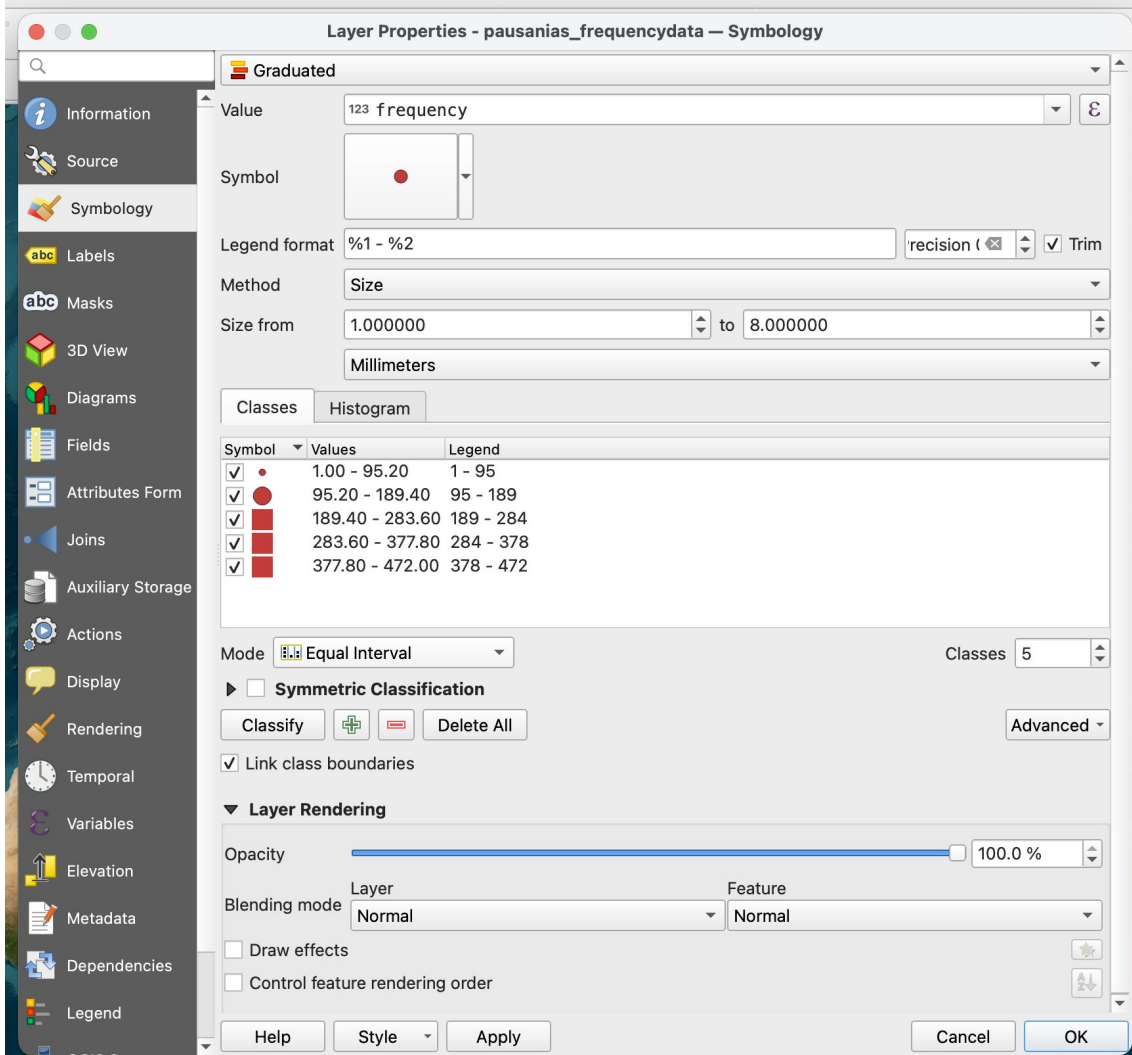
1. Double click on Layer name > Layer Properties
2. Choose styling options for colour, size, labels, etc.



1. Double click on Layer name > Layer Properties
2. Choose styling options for colour, size, labels, etc.
3. Or **Categorize** according to values in the dataset



1. Double click on Layer name > Layer Properties
2. Choose styling options for colour, size, labels, etc.
3. Or **Categorize** according to values in the dataset
4. Or **Graduate** by size or colour if you have numerical data (size, frequency, capacity, etc...)



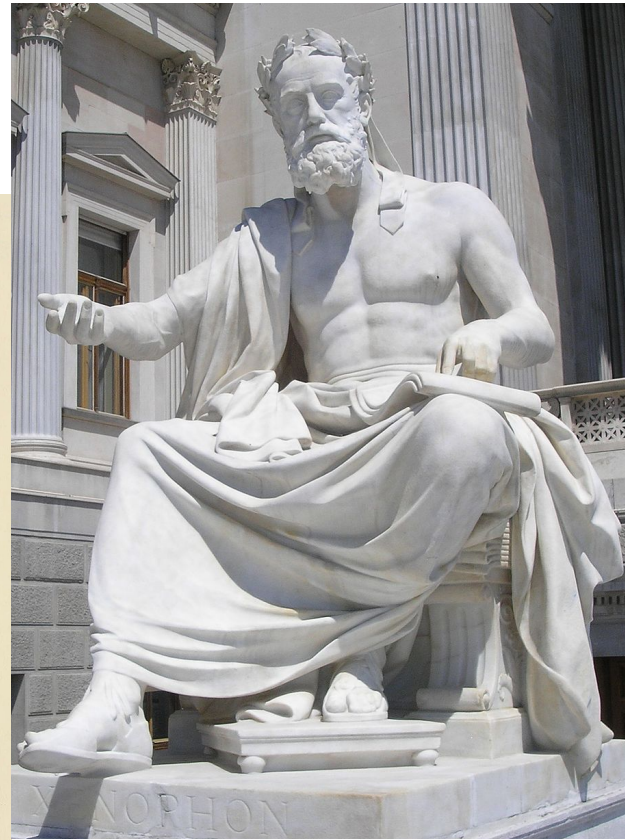
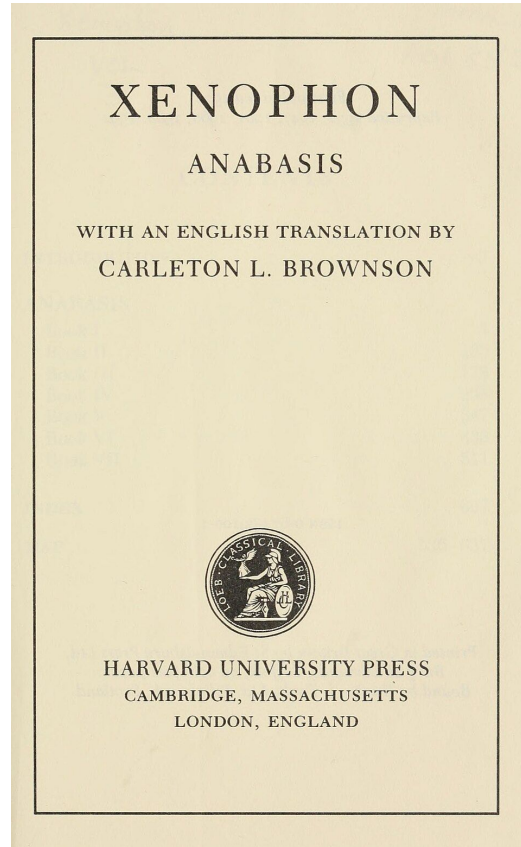
Demo: Least-Cost Analysis in QGIS

ESTABLISHING THE RESEARCH QUESTION



The *Anabasis* of Xenophon

- Military travelogue
- Recounts failed Persian coup from 401-400 BCE, actually written c. 370 BCE.
- *Only* surviving contemporary account of the expedition



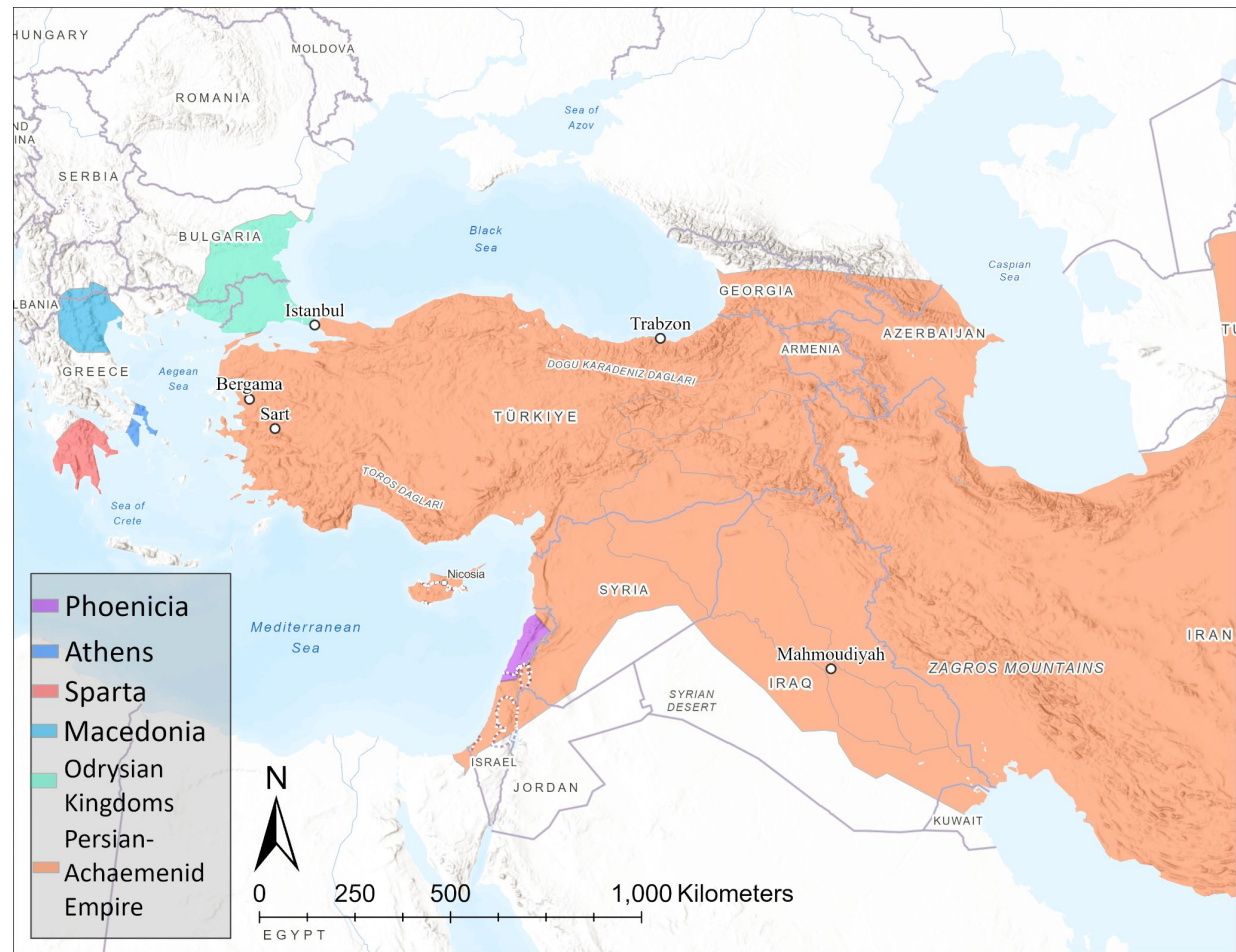
General Route

March:

- Sardis (near modern Sart, Türkiye)
- Battle of Cunaxa (west of modern Mahmoudiyah, Iraq)

Retreat:

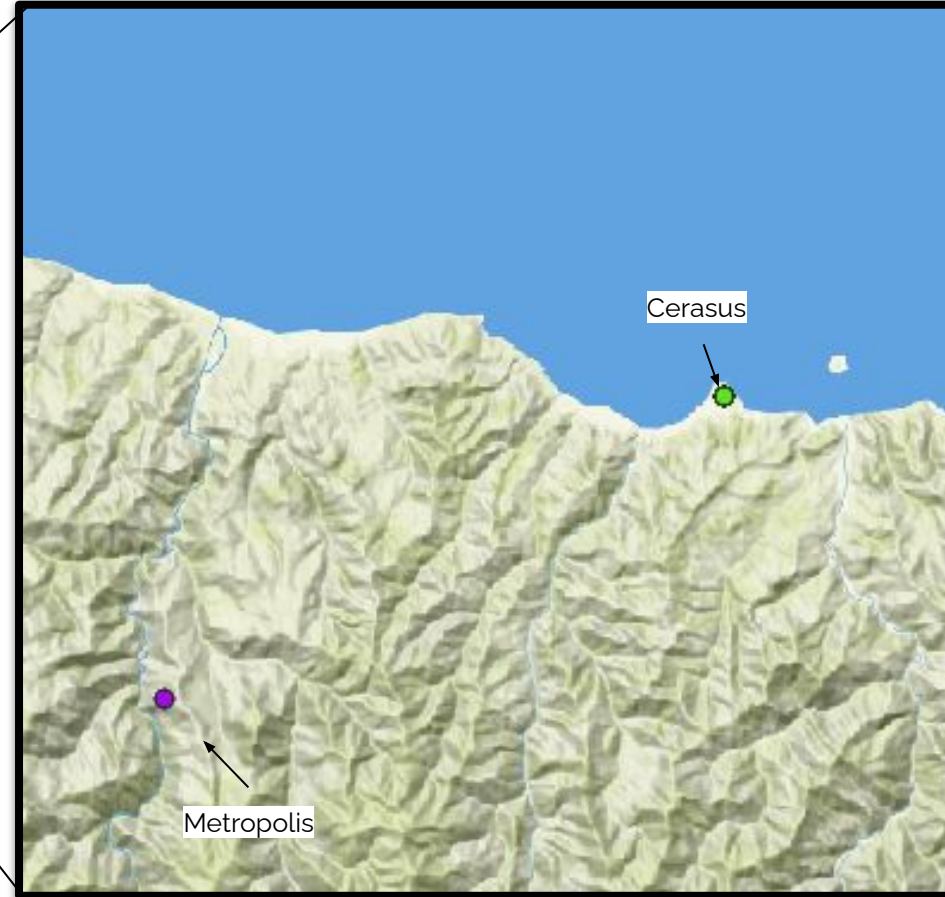
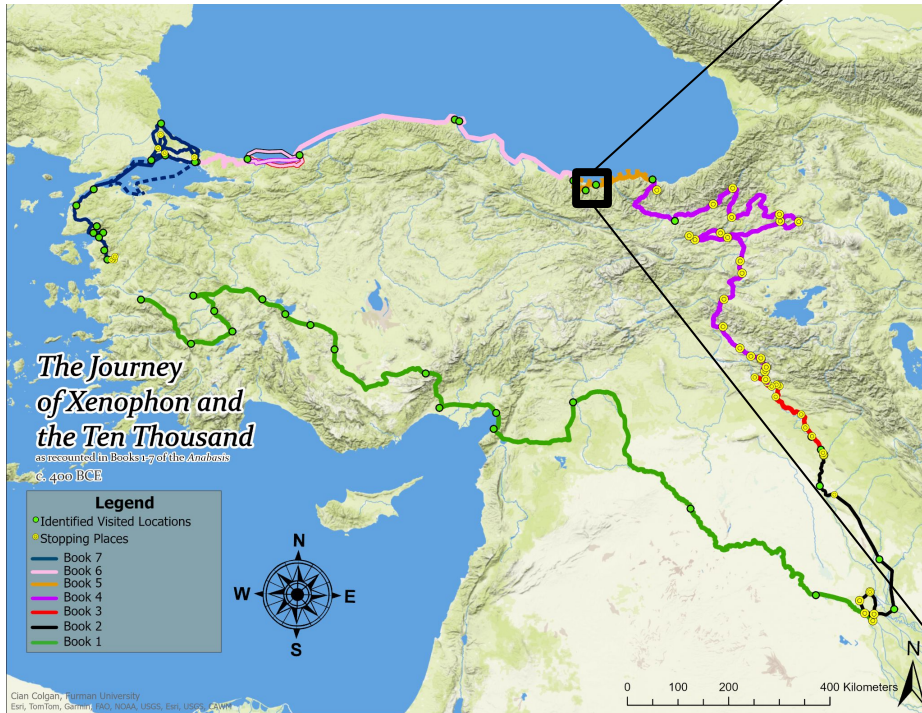
- Trapezus on the Black Sea coast (modern Trabzon, Türkiye)
- Byzantium (modern Istanbul, Türkiye)
- Pergamus (modern Bergama, Türkiye)



General extent of relevant contemporary political entities
(Data modified from AWMC)

Specific Problem: Book 5

Cerasus → Metropolis



DEMONSTRATION: LEAST-COST ANALYSES



Prerequisites

(1)

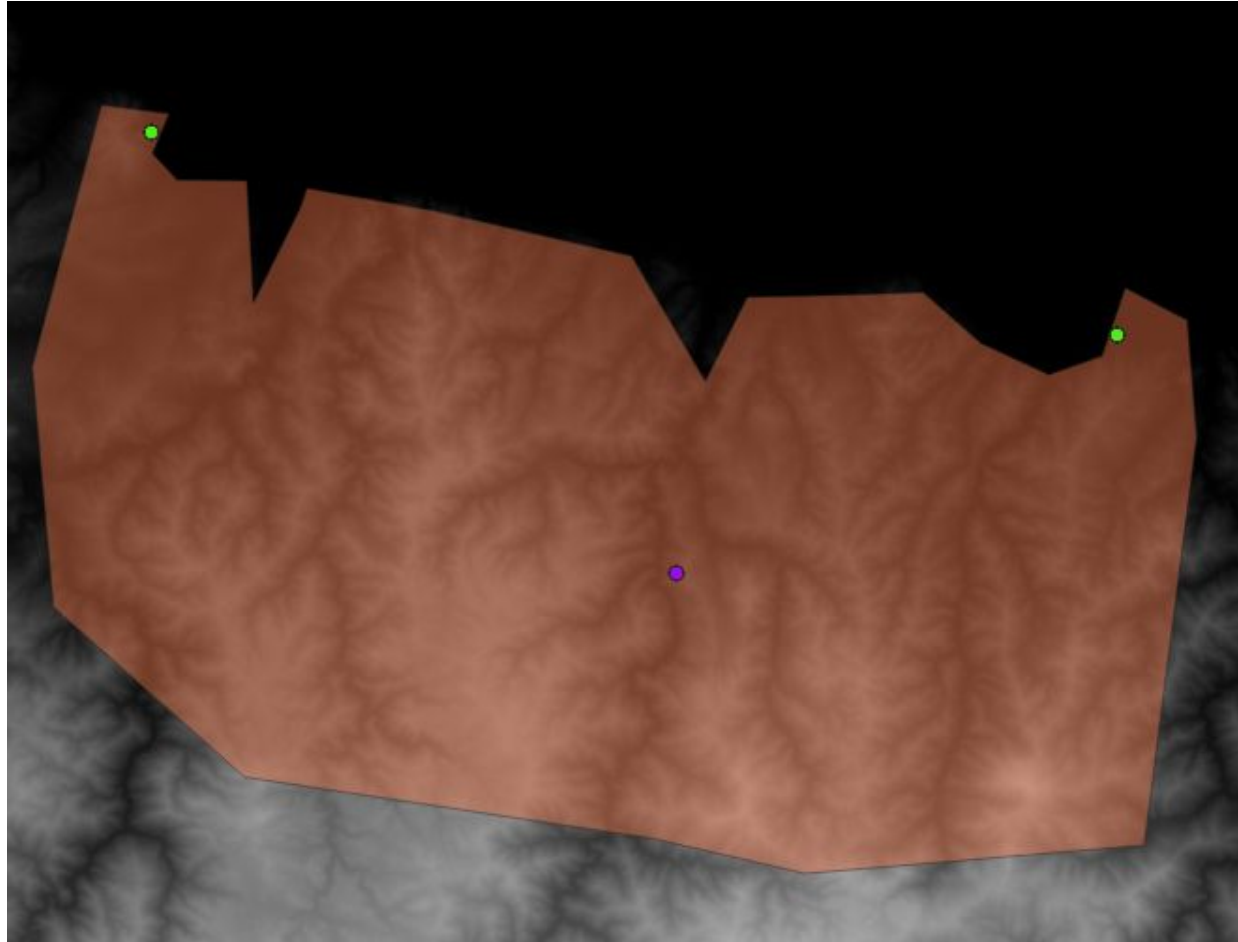
Point data

(2)

Digital elevation
model (DEM)

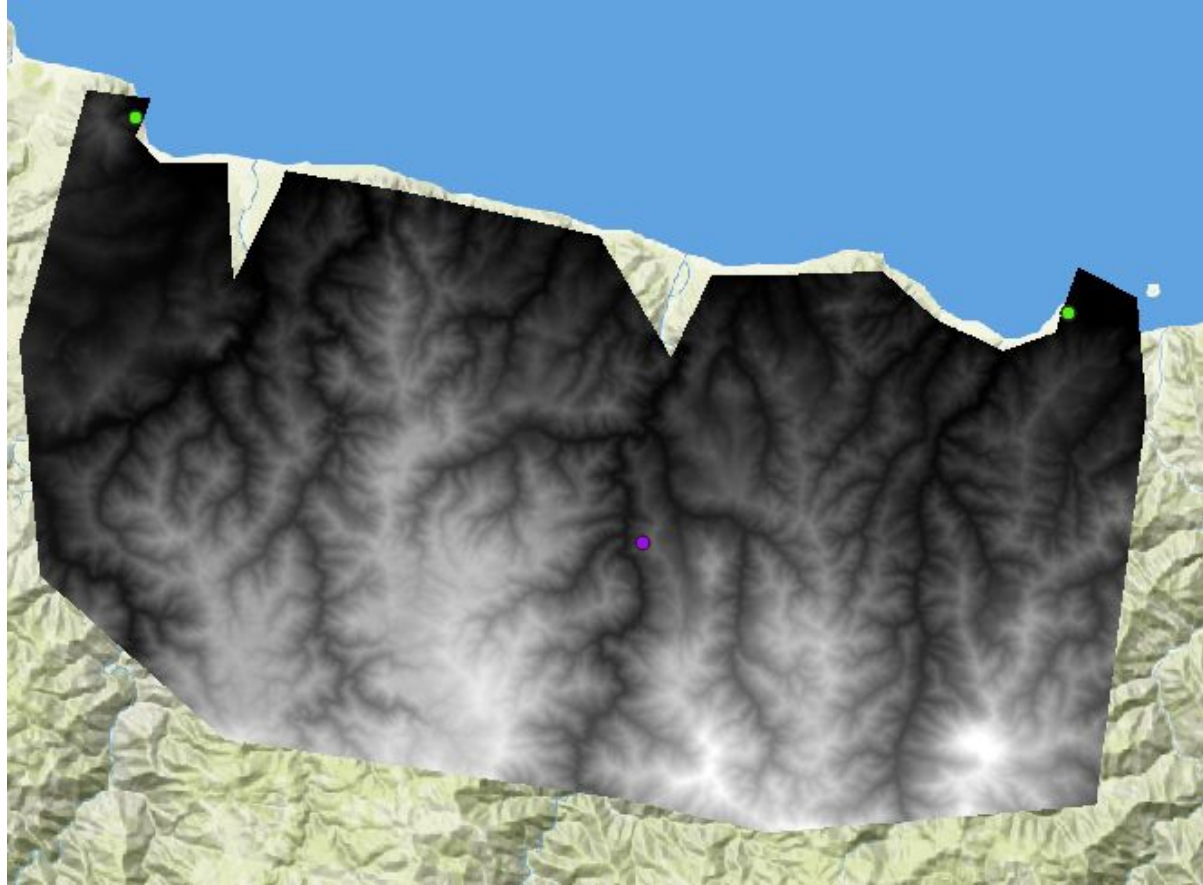
Clipping DEM

1. Layer → Create layer → New Shapefile
2. Toggle edits → Add polygon → Draw vertices



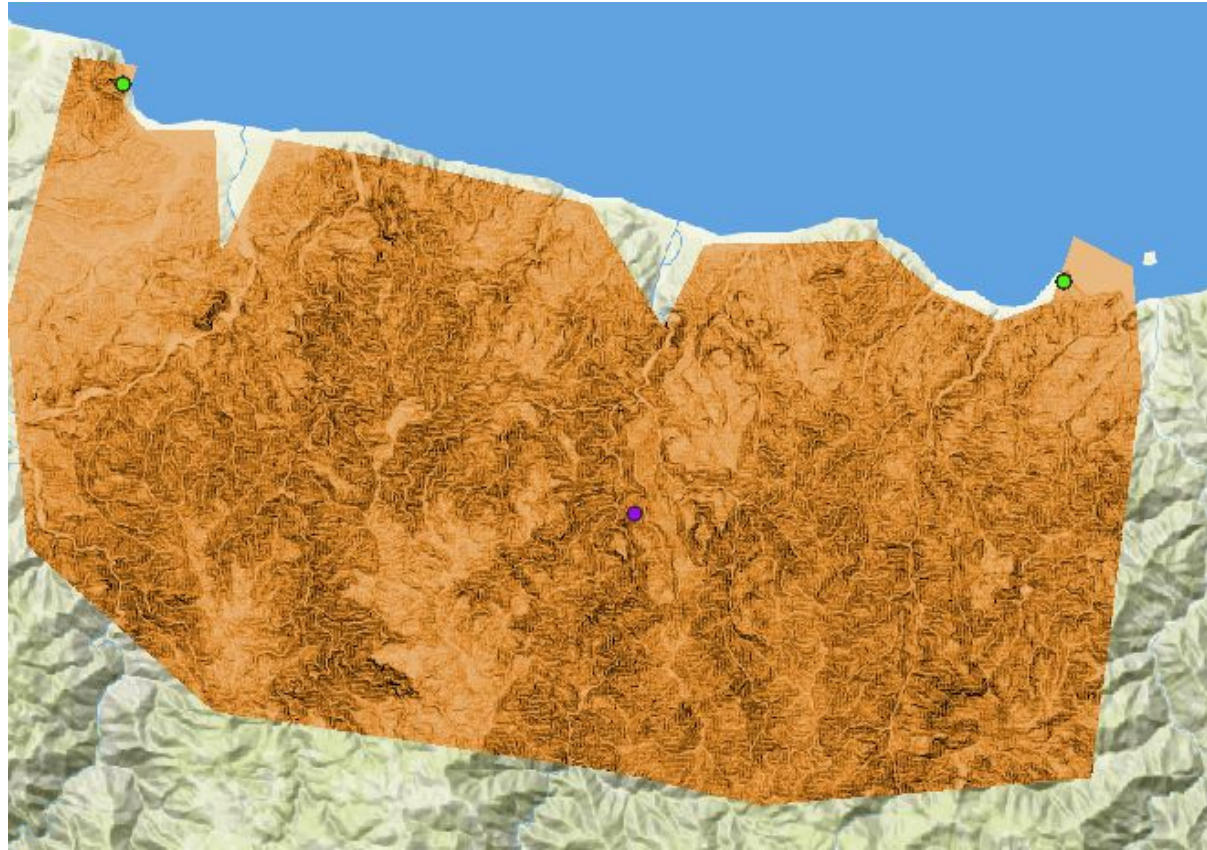
Clipping DEM

1. Raster → Extraction → Clip raster by mask
2. Clip raster by mask:
 - **Input:** DEM
 - **Mask layer:** Polygon
 - **Check box:** clip raster by extent of mask layer
 - Save file



Calculate Slope

1. Raster → Analysis → Slope
2. Slope:
 - **Input:** Clipped DEM
 - Express slope as **percentages**
 - Save file



Adjusted symbology; lighter = less slope (flatter terrain),
darker = more slope (steeper terrain)

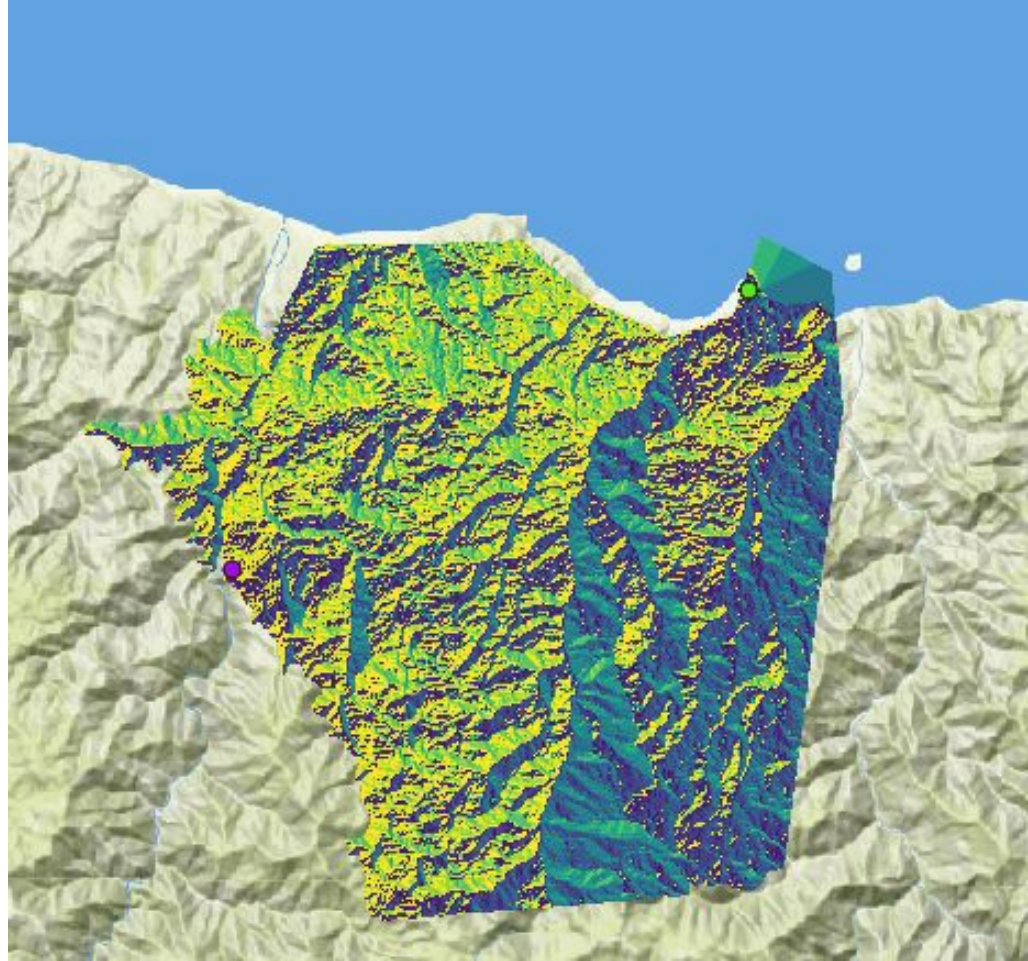
Least-Cost Path

1. Processing → Toolbox → Cost distance analysis → Least Cost Path
2. Least cost path:
 - **Cost raster layer:** Slope
 - **Cost raster band:** Band 1
 - **Start-point layer:** Cerasus
 - **End-point(s) layer:** Metropolis
 - Save file
3. Evaluate!



Calculate Cost Accumulation Rasters

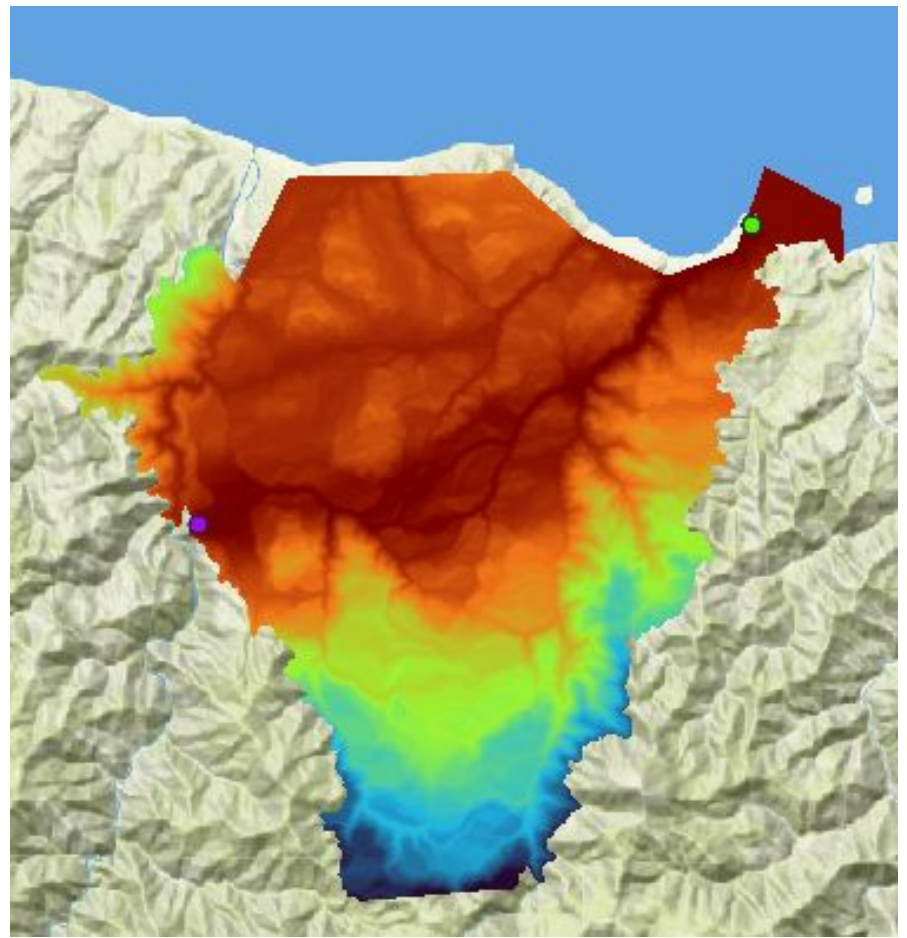
1. Search **r.cost** tool
 - (Processing → Toolbox → GRASS → Raster (r.*) → r.cost)
2. R.cost:
 - **Unit cost layer:** Slope
 - **Starting coords:** Cerasus
 - **End coords:** Metropolis
 - (Optional) Use 'Knight's move'
 - Save all **three** output files
3. Repeat step 2, but swap the starting and ending coords



Movement direction raster overlaying Cost accumulation raster for Cerasus → Metropolis

Combine Cost Accumulations

1. Raster → Raster calculator →
Enter following expression
 - (" [Origin_AccumCost]@1" +
" [Destination_AccumCost]@1")
 - Substitute what is in brackets for your
file names *and remove the brackets []*



Combined accumulation rasters (with adjusted symbology);
dark red = least cost, blue = greatest cost

Combine Cost Accumulations

1. Need to establish percentile threshold
 - 10% is a good rule of thumb
2. *Min value * Desired % threshold*
 - Eg. $16137.36 * 1.10 = 17751.096$
3. Raster → Raster calculator → Enter this expression:
 - `if ("[Combined_AccumCost">@1"
<= [17751.096] ,
"[Combined_AccumCost">@1" , 0/0)`
 - Substitute what is in brackets for your file name and threshold respectively (*and remove the brackets*)

This equation is written to just show the 10% lowest-value pixels, with everything else rendered as a NoValue, which in QGIS defaults to not rendered on the map



Final least-cost corridor output (showing bottom 10% least-cost pixels to traverse). Adjusted symbology; Red = least cost to traverse, blue = greater cost